

Full length article

An ontology-based methodology to establish city information model of digital twin city by merging BIM, GIS and IoT

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ABSTRACT

With the development of digital city and smart city construction, the City Information Model (CIM) has played a critical role as a container of spatial-temporal data to establish the Digital Twin City. For a digital twin city, a virtual high-fidelity CIM model that corresponds closely to the real physical world is the premise and cornerstone of its construction. Therefore, the integration of BIM, GIS and IoT has become the preferred topic for researchers and has received much more attention from a wide academic circle. However, traditional integration mainly focuses on the conversion of both IFC and CityGML, and IoT data are also often used as visualizations. More importantly, the underlying data formats of GIS, BIM and IoT are still independent of each other without a unified data structure expression, so real data-driven analysis and decision-making cannot be implemented. This study aims to establish a general CIM ontology to integrate heterogeneous BIM, GIS and IoT data. First, the related work of BIM, GIS and IoT integration is studied and analyzed. A comparison of three mainstream approaches, data conversion, standard extension and data linking, is conducted, and it illustrates the advantages of ontology techniques in solving data interoperability problems. Second, a technical framework of BIM, GIS and IoT data integration based on ontology technology is proposed. The approach is mainly divided into five steps: geometry processing, data instantiation, ontology construction, ontology mapping and querying application. On the basis of the CIM ontology, an application ontology is built for a specific application domain to illustrate rule-based mapping, querying and inferring. Finally, the case study shows that the Ontology-based methodology in this paper has contributed to establish a general pattern for CIM data integration by mapping and linking concepts from the semantic level. It avoids changes in the original data sources and the missing data problem.

1. Introduction

A city is a huge and complex system with diverse disciplines and a large number of citizens involved. It generates a considerable amount of urban asset management data every day. Accordingly, smart city stakeholders put forward higher requirements for the comprehensive digitalization of cities, namely, a digital twin city, which is a one-to-one corresponding three-dimensional (3D) representation of the real city. The premise and cornerstone of the construction of a digital twin city is a city information model (CIM) that involves spatial information of different scales and daily operation data collected by various IoT (Internet of Things) terminals, such as cameras, temperature sensors, and smoke sensors.

The integration of building information modelling (BIM) and

geographic information system (GIS) can realize spatial information representation and have complementary advantages in the analysis, decision making and management of urban information at the building and city levels. Thus, the integration of BIM and GIS is the basic of the CIM for smart city [1]. On the one hand, the 3D GIS data model avoids maps as intermediary models and focuses primarily on the large-scale outdoor environment. On the other hand, BIM contains rich and detailed structural information in the whole life cycle of urban assets, including indoor rooms and routes, built elements, facilities and equipment. In an integrated environment, BIM architects can design buildings based on the outdoor environment and quickly determine the impact of architectural design on the surroundings. Moreover, BIM models can provide detailed indoor information and directly become a part of the 3D model of urban space. With the addition of building

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Table 1

New data standards for the integration of BIM and GIS.

Name of standard	Organization/Author	Application field	Characteristic/Content
IndoorGML [33]	OGC IndoorGML Standard Working Group	modelling indoor spaces for navigation purposes	A core data model to describe topological connectivity and different contexts; a complementary standard to CityGML, KML, and IFC to support location based services for indoor navigation.
InfraGML [34]	OGC LandInfraDWG	land and civil engineering infrastructure facilities	Areas include facilities, projects, alignment, road, railway, survey, land features, land division, and “wet” infrastructure; enables linking of geospatial and BIM data.
CIM(City Information Model) [36]	Xu X, Ding L, Luo H, et al	case study in campus fire emergency	Reorganize the objects of IFC and CityGML into several modules: building, transportation, city furniture, MEP, water body, etc.
UBM(Unified Building Model) [37]	EI-Mekawy, Ostman, et al	case scenarios in hospital for evacuation and allocating spaces for patient wards	Fusion of IFC and CityGML for building information; manages and visualizes in GIS system
UIM(Urban Information Model) [38]	Clement Mignard, Christophe Nicolle	urban facility management	Defines spatial, temporal and multi-representation concepts; implemented on a self-built platform

materials and energy-related properties, the CIM will become increasingly detailed and practical for life-cycle management and overall city planning. However, in many BIM and GIS integration software tools, the BIM model and 3D GIS data are only fused at the visualization level through location overlap, and the underlying data structures are still independent of each other without a unified data structure expression, so real data-driven analysis and decision making cannot be implemented.

Information sharing between BIM, GIS, and IoT terminals faces many problems and challenges reflected in different users and application fields, different spatial scales, different levels of details, different semantics, and geometric representations [2]. Current studies are mainly based on the transformation and extension of two data standards, IFC (Industry Foundation Class) and CityGML (City Geography Markup Language). IFC is a vendor-neutral and widely accepted data standard proposed by buildingSMART for BIM applications. It is based on EXPRESS, an object-oriented modelling language, and the schema is strictly clear and self-contained. CityGML [3] is an application model of 3D GIS in the field of cities. Based on the GML language, it defines 13 city-related thematic modules and 5 different levels of detail for the expressions of either 2D terrains, bounding boxes, detailed exterior spaces or interior spaces. However, this approach inevitably leads to the loss of information issues, as none of the existing data standards can cover all the elements of the other. It is also time-consuming and laborious to propose a new standard, which needs to establish different standards for different applications. Meanwhile, there are few studies on urban data management, and BIM and GIS data tend to be managed independently [4]. Additionally, the integration of BIM and GIS is still in the early stage of geometric visualization and simple semantic mapping. Failure to grasp the semantics and relationships embedded in the data strongly hinders the effectiveness of urban asset management.

In addition, the daily operation data are always represented in different formats, such as videos, figures, and serialized data. A comparison between BIM and the Internet of Things reveals that BIM and IOT differ greatly in development focus, application area, reference system and data storage. BIM is mainly aimed at the internal environment of the building. By modeling every detail of the building, including the structure, appearance, attributes, etc., it can also realize the functions of structure analysis, construction cost estimation and building maintenance. IOT integrates real-time data such as building geographic coordinates, indoor temperature, indoor oxygen and other gas levels, sound level, human and animal presence, humidity, building component status (i.e. window opening/closing), elevator and escalator status (i.e. whether running), air pollution, noise level, solar radiation level, parking space occupancy, bus location, etc. Underground location, etc. The data that need real-time transmission of sensors on site is data that can be provided by IOT and can not be provided by BIM at present. BIM

data is data that will not change for a period of time, while IOT data is data that changes constantly in real-time. For example, the material of building interior doors may not change in a year or two, and the switch state of doors may change several times in a hour or two [5]. At the same time, building coordinates in Internet of Things information need to be matched with building coordinate calibration in BIM information. BIM usually uses local plane coordinate system because building size is limited and space range is relatively small. IOT represents the use of 2D or 3D geographic information to simulate all types of natural and human products in cities, countries and the world, so the Earth's sphere needs to be considered when the Internet of Things uses coordinates. Because of the different concepts behind BIM and the Internet of Things, they use different data structures to store and exchange data. The Internet of Things and BIM use different methods to represent 3D geometries. An IFC file represents a 3D geometry using one or a combination of CSG, sweep volume and boundary representations. The Internet of Things usually uses only boundary representations, such as CityGML and polyhedron representations [6]. BIM information mainly refers to engineering, while the Internet of Things application projects are various, including industry, logistics, park, medical treatment, etc. It is difficult to directly include IOT data and BIM information by adopting a unified standard. Therefore, an integrated integration method of BIM and Internet of Things is urgently needed [7,8].

Semantic web technologies are considered the most effective and promising method [9] to achieve the interoperability and integrated management of semantic information and knowledge in the urban space. With the natural ability to incorporate information from different data sources, the semantic web aims to construct conceptual ontologies in the form of annotations to provide machine-accessible semantics [10] and link heterogeneous data from the upper-layer concepts and logics.

The term “Semantic Web” is presented by World Wide Web Consortium (W3C), which is an international community dedicating itself on developing Web standards [11]. It refers to W3C's vision of the Web of Data, which extends the concept of Web of Document by adding the metadata to the separate data and linking them with expressive relations. To be noticed, Sir Tim Berners-Lee, the inventor of World Wide Web and the leader of W3C, emphasized the Semantic Web is about making links rather than just putting data on the web, so that a person or machine can explore the web of data to find the other, related data [12]. As a result, computers will be intelligent enough to “understand” the data of interest similar to the way humans process information and support trusted interactions.

Ontology is a fundamental concept of Semantic Web. According to W3C, it refers to the concepts and relationships used to describe and represent an area of concern [13]. Another similar but clearer definition from Ontotext is that an ontology is a formal description of knowledge as a set of concepts within a domain and the relationships that hold

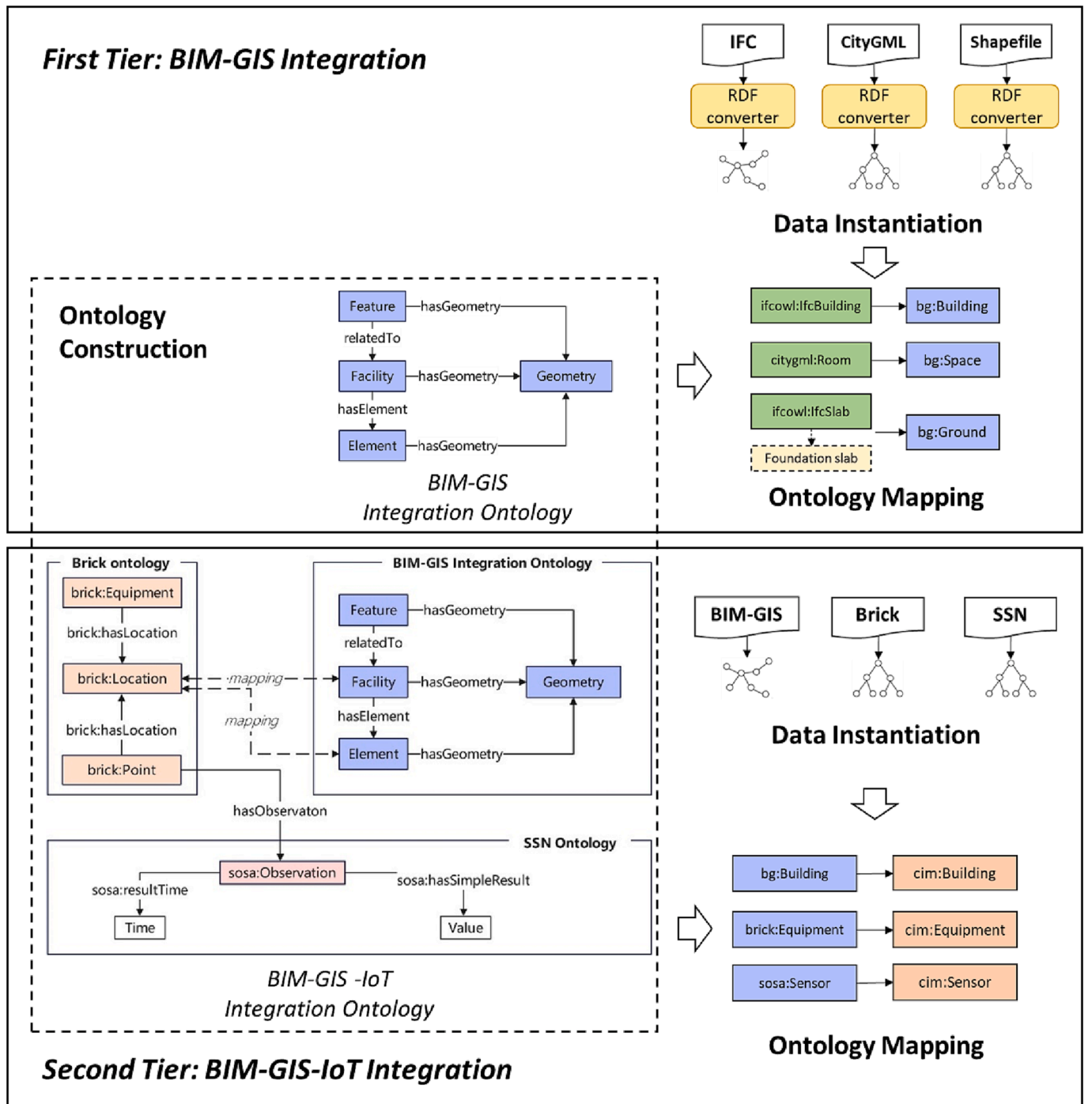


Fig. 1. A tiered approach of BIM, GIS and IoT integration based on ontology technology.

between them [14]. Therefore, ontologies serve as the basic building blocks for inference techniques on the Semantic Web. Due to the nature of Ontology to describe the domain of concern in a semantically related way, it is universally adopted in various fields, including AEC [15–18], Geography [19–21], Government [22], Life Sciences [23], Linguistics [24], Media [25] and etc. At the same time, ontology becomes a prevalent choice of integrating the concepts of different domains and establish the pertinent relationships between them [26,27].

To make the Web of Data a reality, W3C suggests having the huge amount of data on the Web available in a standard format, reachable and manageable by Semantic Web tools. Resource Description Framework

(RDF) is, hence, proposed as a common standard for the data interchange of the highly interconnected data. It empowers effective data integration from various sources, detaching data from its schema. The simple expression of RDF makes it popular in practice. Following the previous work of ontology engineering, this research will adopt RDF to represent the formal description of the proposed integrated ontology, CIM.

Ontology provides the formal expression (vocabulary) of the concepts and their relationships that are commonly accepted in a specific domain, thus eliminating the semantic bias of the same entity in different data structures [28]. To date, as the integration of BIM-GIS

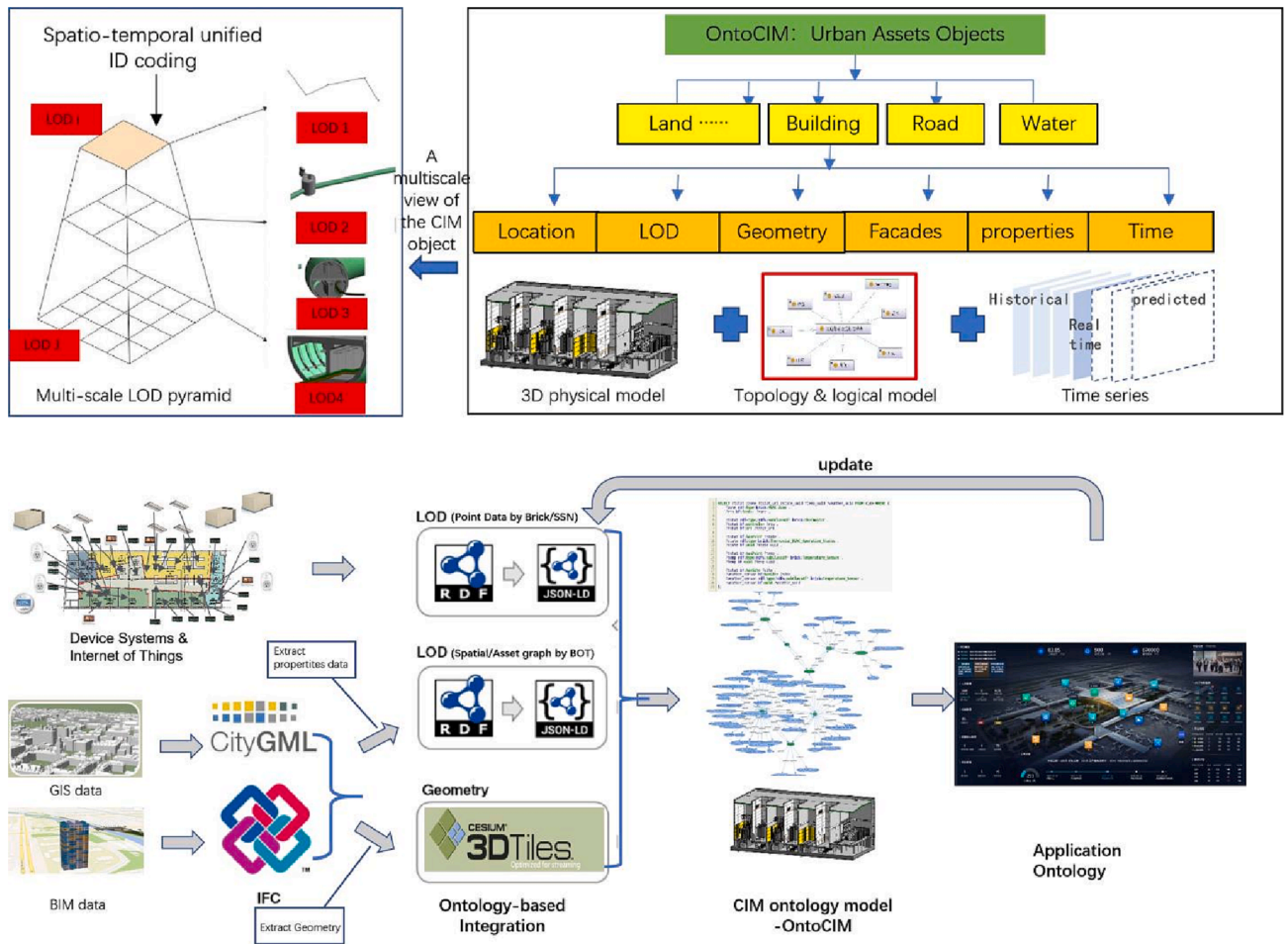


Fig. 2. A CIM ontology framework model (OntoCIM) Pattern and Processing Flow.

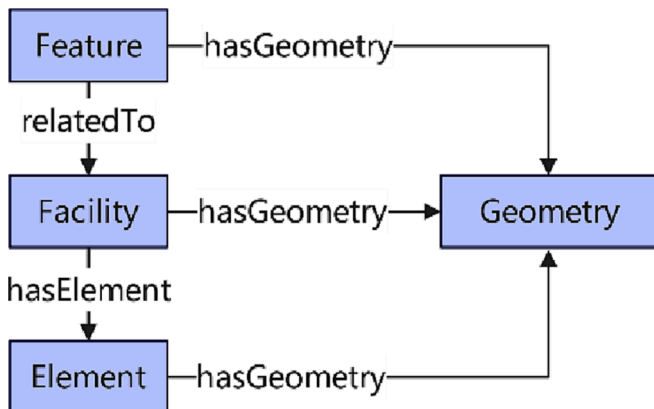


Fig. 3. Linkages between the top main classes of the BIM-GIS Integration Ontology.

depicts, there is also a trend to develop ontology-based approaches to fully support the semantic description of those data and further big data analysis. Among those approaches, Brick together with Semantic Sensor Network Ontology (SSN) is one of the most prevalent since it has the powerful capability to describe physical, logical and virtual service facilities of urban assets, the relationships between them and their dynamic characteristics.

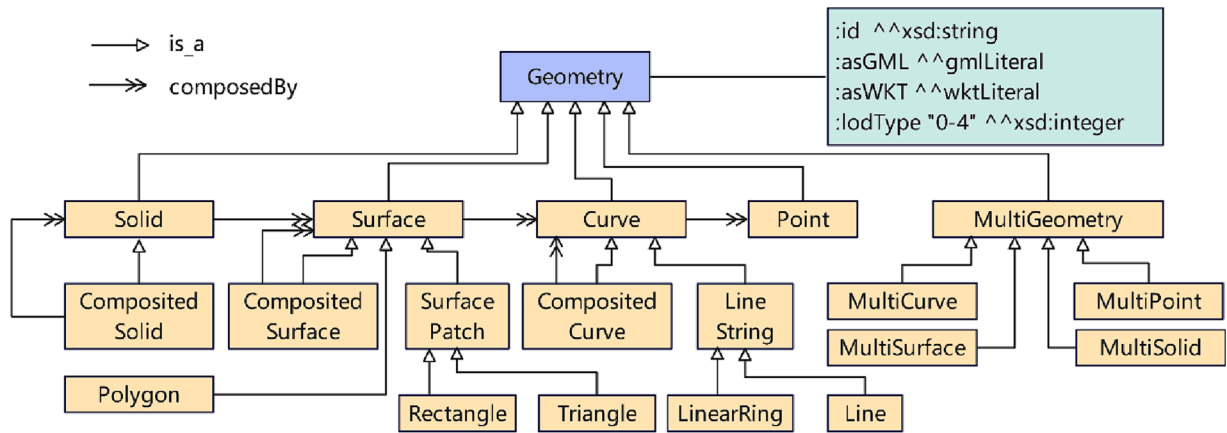
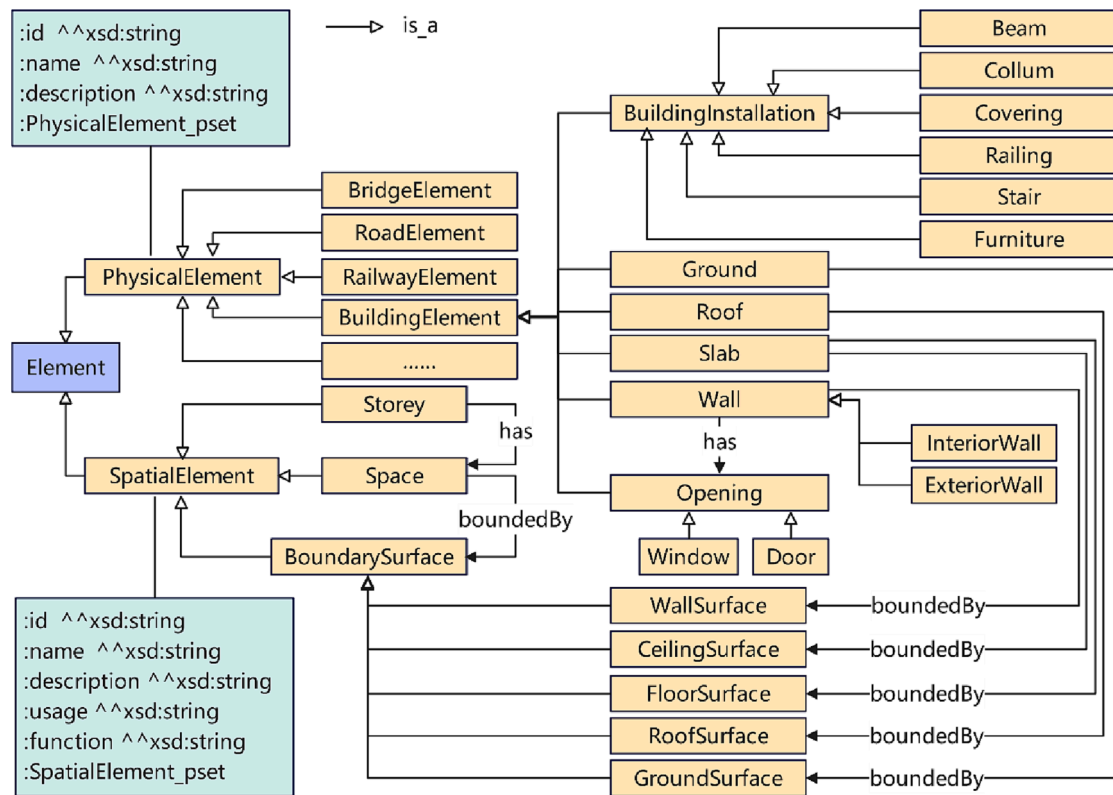
Therefore, this study aims to establish a general CIM ontology to

integrate heterogeneous BIM, GIS and IoT data. The second section of this paper mainly studies the related work of BIM, GIS and IoT integration. By comparing the three kinds of methods, data conversion, standard extension and data linking, it shows the advantages of ontology technology in solving the problem of data interoperability. Section 3 proposes the technical framework of BIM, GIS and IoT data integration based on ontology technology. The method is mainly divided into five steps: geometry processing, data instantiation, ontology construction, ontology mapping and querying application. On the basis of the CIM ontology, an application ontology is built for a specific application domain in section 4 to illustrate rule-based mapping, querying and inferring. Finally, the last section summarizes the achievements and contributions of this paper and puts forward the shortcomings and prospects.

2. Background and related work

2.1. Data conversion and extension to achieve integration of BIM and GIS

With regard to the integration of BIM and GIS, some studies have focused on data conversion and extension. Donkers [29] proposed an automatic conversion method from the IFC model to CityGML LoD3 (LoD, Level of Detail) based on semantic mapping and geometric calculation. Some commercial software such as FME [30] and IFCExplorer [31] can also realize conversion of BIM and GIS, but the results do not fully conform to the CityGML specification and have problems of data error and missing, as IFC generally have much more

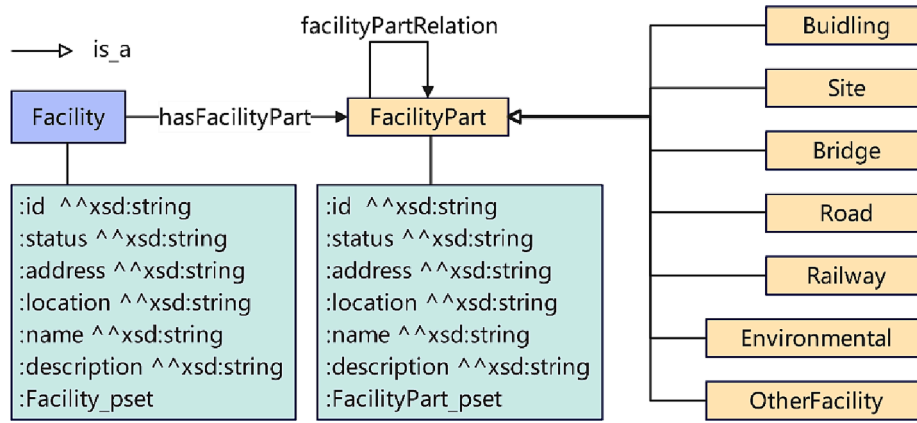
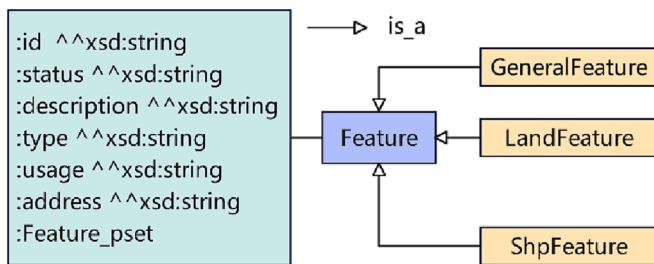
Fig. 4. Hierarchy structure of *Geometric* (partly).Fig. 5. Hierarchy structure of *Element* (partly).

information than CityGML. Geometry must be harmonized with semantics to transform from solid to surface and to assign different semantics to different surfaces [1]. Such a process is very complicated and error prone.

Except for the data conversion, data extension denotes the expansion of the existing standards or the introduction of new standards. CityGML provides application domain extension (ADE), which allows definitions of new city objects and attributes at the data schema level. GeoBIM is a representative project [32] that uses CityGML ADE to convert CAD/BIM data into GIS identifiable construction information. This method is applicable for specific use and requires a large amount of schema modification. New standard development usually involves union operations of part of IFC and CityGML for specific applications. Table 1

shows some new data standards based on IFC and CityGML. The Open Geospatial Consortium (OGC) has made great efforts to promote the exchange and interoperability of geospatial data [33,34]. Although the open standards provide rich classes, self-contained relations and hierarchical structures, they are based on modelling languages such as XML, GML and UML and limited to the level of syntax and structure [35]. Meanwhile, it is not only difficult to modify the standards but also difficult to make them widely accepted and applied. Therefore, handling the interoperability problem from both syntax level and the semantic level is expected to fundamentally solve the problem.

According to proposal by Kang et al. [39], some groups, about studies on BIM and GIS data integration, are divided as follows: (1). Schema-based approaches; (2). Service-based approaches; (3). Process-

Fig. 6. Hierarchy structure of *Facility* (partly).Fig. 7. Hierarchy structure of *Feature* (partly).

2.2. Ontology-based approach to achieve integration of BIM and GIS

Another popular scheme to achieve the integration of BIM and GIS is the ontology-based approach. Compared with modification of the original data model, ontology represents concepts in the real world rather than a human-defined framework or expression, so ontology is more suitable to meet the needs of data exchange and knowledge sharing for the user community. The ontology-based approach has shown its potential in dealing with the interoperability issue in engineering industry, such as the integration of building information model and building codes [40], seamless information exchange during production life-cycle [41], and multi-data source integration [42]. Moreover, in the architecture, engineering, and construction (AEC) industry, a series of studies have laid the foundation of the ontology-based approach for integration of BIM and GIS. On the one hand, BIM researchers are increasingly interested in the “semantics” of buildings. Pan et al. [43] proposed the use of semantic web technology in the AEC industry in 2004. Rezgui et al. [44] proposed enhancing semantics through Ontology to make up for the shortcomings of IFC product data. The idea is that ontologies do not attempt to replace or alter a single data model but instead act as a layer of semantic abstraction on top of current standards and models, focusing on overlapping multiple domains and specific concepts to further integrate data in a smarter way. BuildingSMART’s LDWG working group identified ifcOWL (by Pauwels and Terkaj [16]) as open domain ontologies, which strictly follow the EXPRESS schema of IFC.

On the other hand, geographical ontology was proposed as early as the middle and late 1990 s. Bittner and Stell [45] and Frank [21]

based approaches. (4). System-based approaches. However, these modeling approaches for data integration are mostly done manually, meaning that it requires a lot of manual effort and time to complete. Ontology-based approaches are a type of data integration approach that is used to integrate heterogeneous data models. This approach involves extracting information from different data models and then generating a common ontology model using a general ontology model, such as the resource description framework (RDF). RDF is a standard model for representing information on the web, and it is used to represent data in a structured way. Once the information has been extracted and the common ontology model has been generated, data mapping rules are applied between the two heterogeneous models using ontology operators.

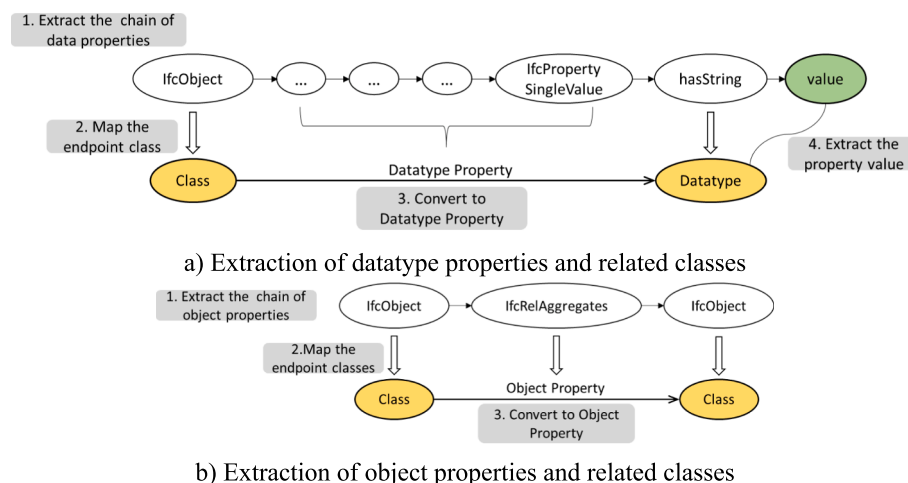
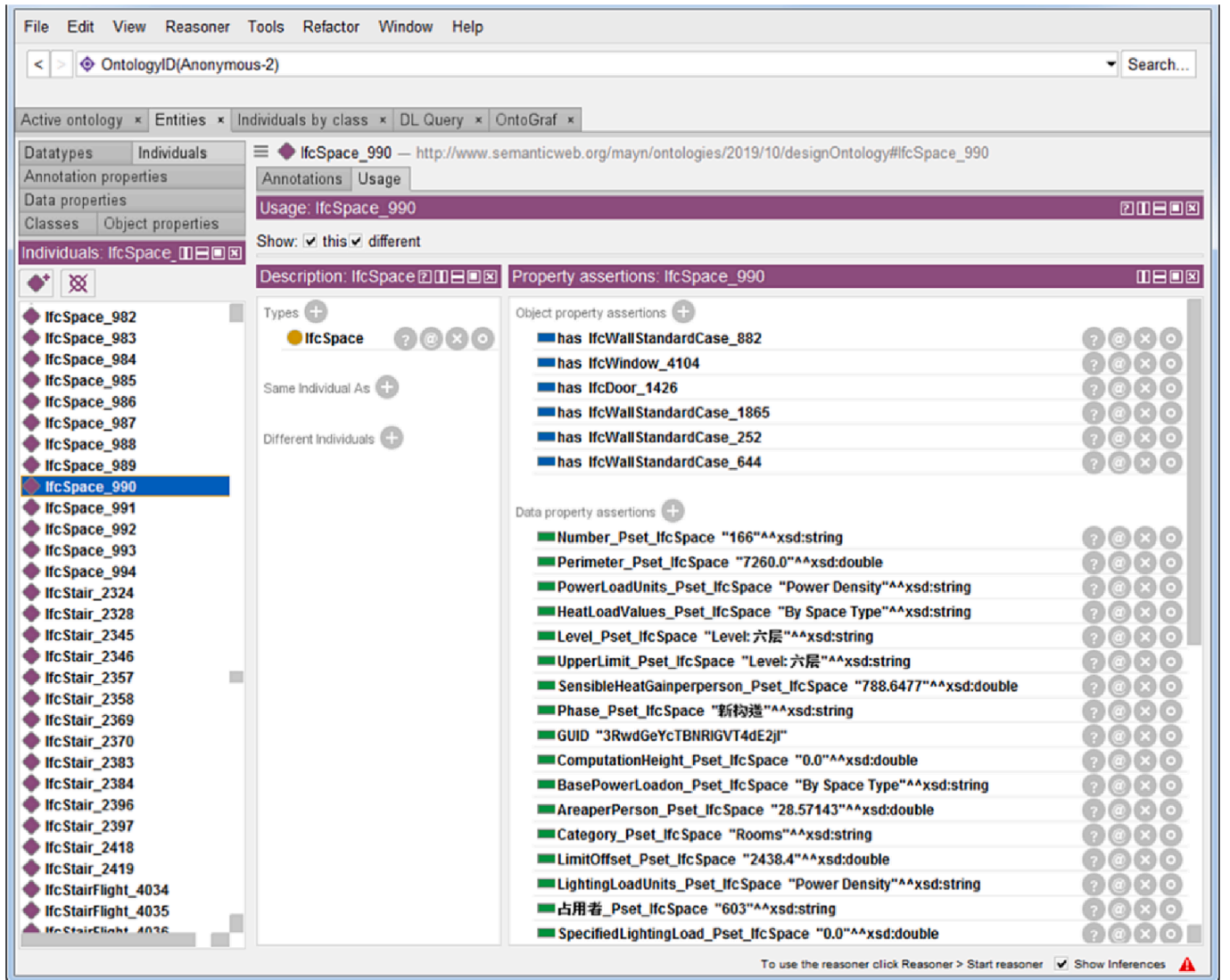


Fig. 8. Automatic extraction and simplification of ifcOWL.

Table 2

Examples of SPARQL queries used in this paper.

No.	Content	SPARQL query
SQ1	Obtain the attribute name and related value of IFC objects	<code>SELECT?obj?n?value WHERE {?rel ifcowl:relatedObjects_IfcRelDefines?obj.?rel rdf:type ifcowl:IfcRelDefinesByProperties.?rel ifcowl:relatingPropertyDefinition_IfcRelDefinesByProperties?pset.?pset ifcowl:hasProperties_IfcPropertySet?p.p ifcowl:name_IfcProperty?id.?id ifcowl:hasString?n.?p ifcowl:nominalValue_IfcPropertySingleValue?va.?va express:hasString?value. }</code>
SQ2	Obtain relations between IfcSite, IfcBuilding, IfcBuildingstorey and IfcSpace	<code>SELECT?a?b WHERE {?rel ifcowl:relatingObject_IfcRelDecomposes?a.?rel ifcowl:relatedObjects_IfcRelDecomposes?b. }</code>
SQ3	Obtain IfcSpace and related IFC objects	<code>SELECT?room?element WHERE {?boun ifcowl:relatingSpace_IfcRelSpaceBoundary?room.?boun ifcowl:relatedBuildingElement_IfcRelSpaceBoundary?element. }</code>
SQ4	Obtain IfcBuildingstorey and related IFC objects	<code>SELECT?sto?ele WHERE {?rel ifcowl:relatingStructure_IfcRelContainedInSpatialStructure?sto.?rel ifcowl:relatedElements_IfcRelContainedInSpatialStructure?ele. }</code>
SQ5	Obtain IfcWallStandardCase and related IfcWindow/IfcDoor	<code>SELECT?wall?dw WHERE {?relvoid ifcowl:relatedOpeningElement_IfcRelVoidsElement?op.?relvoid ifcowl:relatingBuildingElement_IfcRelVoidsElement?wall.?relfill ifcowl:relatingOpeningElement_IfcRelFillsElement?op.?relfill ifcowl:relatedBuildingElement_IfcRelFillsElement?dw. }</code>

**Fig. 9.** Results of IFC instantiation.

conducted basic research on relevant theories of geographical ontology. In terms of ontology application research, Fonseca [46] has performed much research on ontology-based geographic information integration. He believes that ontology is semantically richer than a database conceptual model, and therefore, it is closer to users' conceptual cognition of the real world. At the same time, a mapping framework between spatial ontology and geographic concept schema is proposed, and it is

suggested to abstract ontology concepts from three levels: formal level, domain level and application level. The emphasis of geographic ontology and traditional GIS is quite different, as traditional GIS is more focused on the spatial and temporal analysis of geographic data, and geographic ontology emphasizes the formal expression of geographical concepts and knowledge to promote semantic interoperability and geographic information sharing. This is very similar to the purpose of

Table 3
IFC objects corresponding to LoD0-4.

IFC Objects	LoD0	LoD1	LoD2	LoD3	LoD4
IfcBuilding	✓	✓	✓	✓	✓
IfcWall (Exterior wall)			✓	✓	✓
IfcRoof			✓	✓	✓
IfcOpeningElement				✓	✓
IfcDoor				✓	✓
IfcWindow				✓	✓
IfcSpace					✓
IfcWall (Interior wall)					✓
IfcSlab					✓
IfcBeam					✓
IfcColumn					✓
IfcStair					✓
IfcRailing					✓
IfcRamp					✓
IfcFurnishingElement					✓

applying Ontology in the BIM domain and fully illustrates the significant advantages of Semantic Web and Ontology. In terms of research on geospatial data sharing, many studies on geospatial semantics have been carried out. GeoNames [19] published ontologies for sharing global toponymy information. Stadler et al. [47] have converted OpenStreetMap data into LinkedGeoData, a large geographic spatial knowledge base shared by the Semantic Web. In addition, the W3C has

published a series of geographic shared vocabularies. The WGS84 lat/long vocabulary [48] uses precision (geo:lat), latitude (geo:long) and elevation (geo:Alt) to describe the geographic location of points in the WGS84 coordinate system. GeoSPARQL [20] defines the core ontology vocabulary, which is used to express the geometry of features and spatial topology, and defines special geometric expressions to support queries based on spatial semantics.

On these basics, there have been many applications of ontology-based approaches for the integration of BIM and GIS. One approach is to use ontologies as intermediation to represent similar concepts in different data models. Deng et al. [49] used reference ontology to integrate IFC and CityGML, including geometric transformation, semantic mapping, and coordination of different LoDs in CityGML. Karan et al. [50] proposed more common detailed methods and workflows to ensure semantic interoperability that included five steps: conceptualization, integration, formalization, manipulation and validation. In this way, ontology is still treated as a tool for information exchange between existing BIM and GIS tools.

Another idea is to build ontology for some specific applications in construction engineering and urban design, but the methods of ontology construction are distinct. Some studies have proposed methodologies that link contents from different data sources. Hor et al. [51,52] integrated GIS and BIM by linking similar concepts and relationships between resource definition framework (RDF) graphs constructed from the IFC model and GIS model. Tuyen Le and H. David Jeong [35] presented the interconnection of Ontology-based data to support decision making

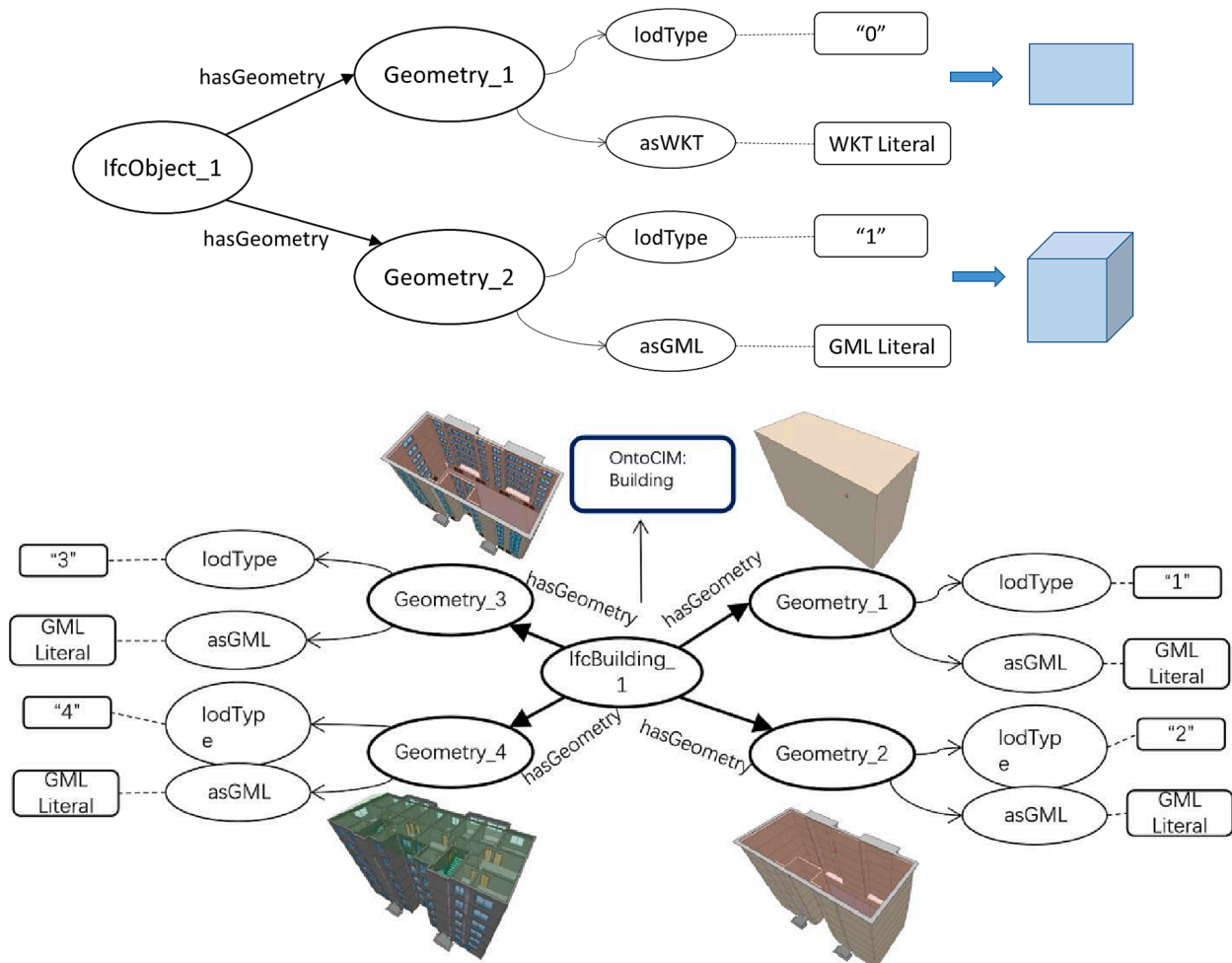


Fig. 10. The LOD Mapping and organization of geometric information based on CIM Ontology.

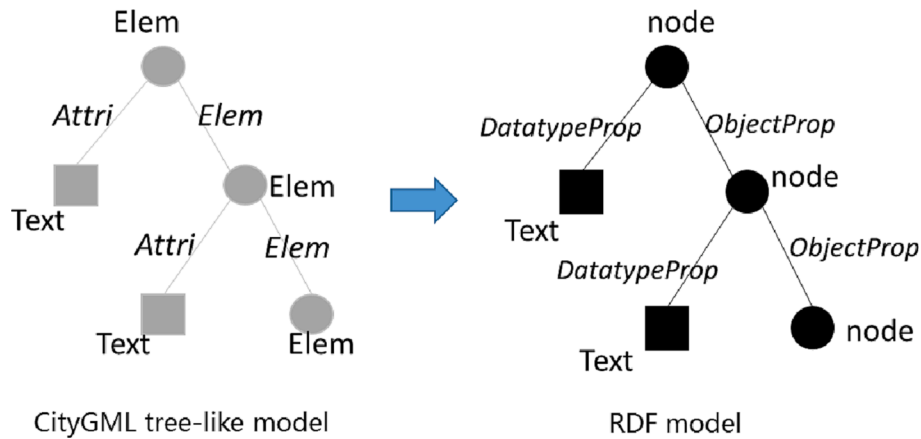


Fig. 11. Comparison of CityGML and the RDF model.

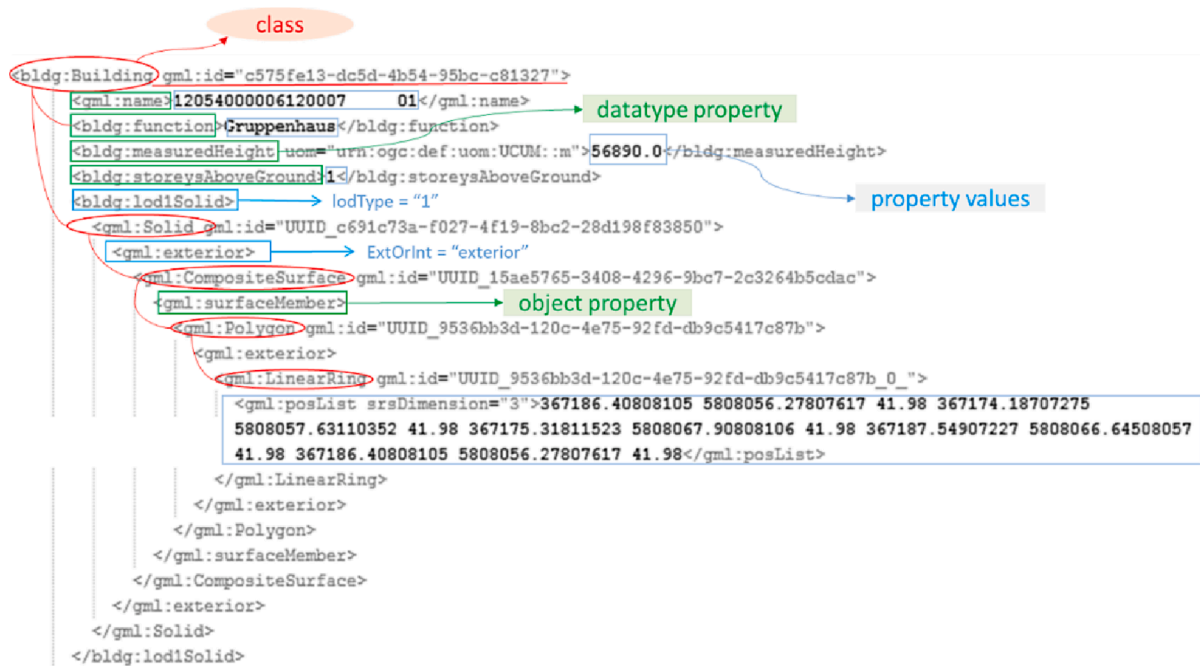


Fig. 12. Mapping from the CityGML file to RDF.

in highway asset management. Karan and Irizarry [53] presented BIM and GIS data as linked data to manage preconstruction operations. The data-linking mechanism is efficient and practical for specific uses, but inconsistencies between the ontologies still exist, especially for BIM and GIS ontologies, because they might have different levels of granularity [50]. Comparatively, some systematic ontologies have also been proposed. The urban information model (UIM) [38] defined the concepts of time, space and multiple representations for urban infrastructure management. Montenegro et al. [54] automatically transformed the land classification standard (LBCS) into ontology and described five different aspects of urban space: activity, function, structure, ownership and site, which were used as the basic structure of the CIM model to support data sharing and reuse of urban planning. Daneshfar et al. [55] designed an urban ontology to represent surrounding information of buildings. The proposed ontology provides a shared knowledge basic for building renovation planning application and for the integration of BIM and GIS.

Obviously, a systematic ontology is more conducive to sharing urban data and multiple application scenarios. However, if an ontology

architecture is too complex, it defeats its purpose and becomes a different form of “data model.” Therefore, this study aims to establish a relatively simple and general urban data ontology and to study the methodology of integrating heterogeneous BIM-GIS-IoT data.

2.3. The integration of BIM, GIS, and IoT

In comparison with other man-made assets, most infrastructure assets are designed for a relatively long life [56]. To improve the efficiency of urban asset management, it is necessary to manage both the static and dynamic attributes of assets in a tractable, easy-to-update way. In addition, when tackling large-scale and linear infrastructures such as roads, railways, bridges, and power grids, geographic factors cannot be overlooked. In recent years, quite a few researchers have recognized the benefits of integrating BIM, GIS and IoT during the O&M (Operation and Maintenance) stage of building or infrastructure projects.

The corresponding design pattern and preliminary implementation have been studied in various subdomains, such as energy consumption

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@prefix De: <http://www.semanticweb.org/mayn/ontologies/2019/10/designOntology#> .
@prefix owl: <http://www.w3.org/2002/07/owl#> .
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .
@prefix xml: <http://www.w3.org/XML/1998/namespace> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

cg: Building_c575fe13-dc5d-4b54-95bc-c81327 a cg: Building;
cg: name "12054000006120007 01"^^xsd:string;
cg: function "Gruppenhaus"^^xsd:string;
cg: measuredHeight "56890.0m"^^xsd:string;
cg: storeyAboveGround "1"^^xsd:integer;
cg: hasGeometry cg: Solid_UUID_c691c73a-f027-4f19-8bc2-28d198f83850 .

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cg: lodType "1"^^xsd:integer;
cg: ExtOrInt "exterior"^^xsd:string
cg: composedBy cg: CompositeSurface_UUID_15ae5765-3408-4296-9bc7-2c3264b5cdac .

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cg: composedBy cg: Polygon_UUID_9536bb3d-120c-4e75-92fd-db9c5417c87b .

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cg: composedBy cg: LinearRing_UUID_9536bb3d-120c-4e75-92fd-db9c5417c87b_0 .

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cg: asGML "<gml:posList srsDimension='3'>367186.40808105 5808056.27807617 41.98 367174.18707275
5808057.63110352 41.98 367175.31811523 5808067.90808106 41.98 367187.54907227 5808066.64508057 41.98
367186.40808105 5808056.27807617 41.98</gml:posList>"^^cg: gmlLiteral .

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Fig. 13. Result of conversion from CityGML to RDF.

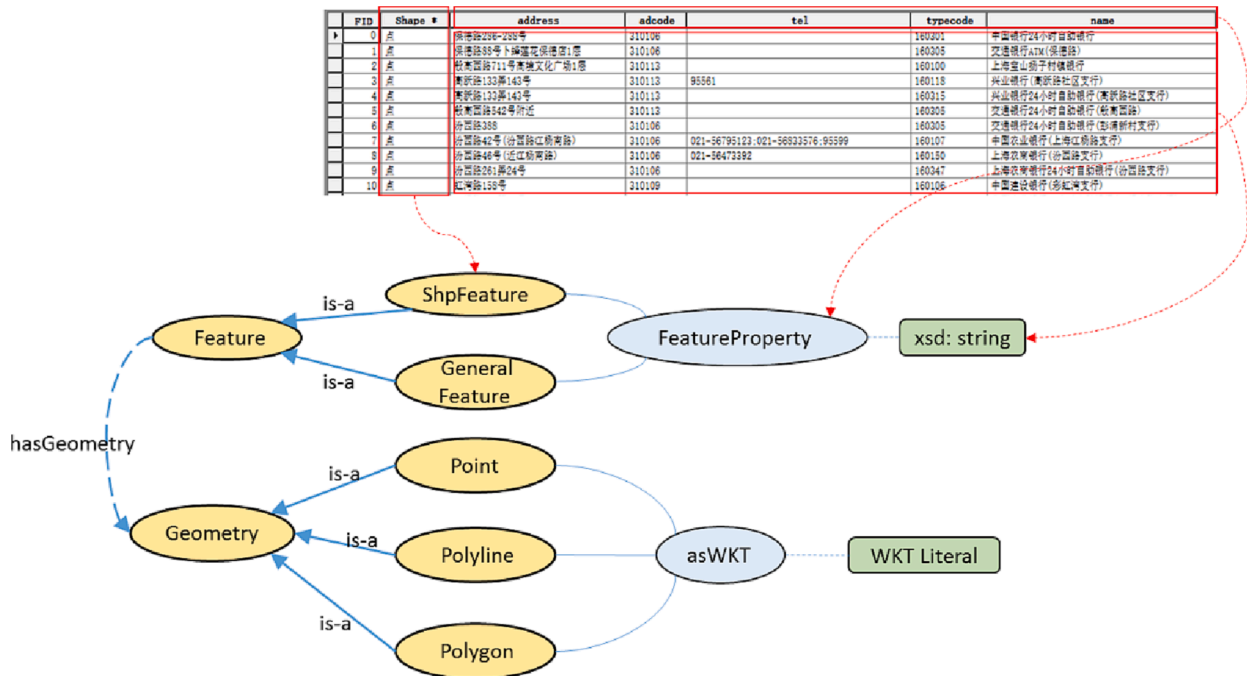


Fig. 14. Instantiation of Shapefile objects.

management [57,58], general urban management [5], risk management [51,59,60], and construction management [61–63]. Bottaccioli et al. [57] proposed a software architecture for the management and simulation of energy behaviors in buildings that integrates heterogeneous data such as BIM, IoT, GIS (Geographical Information System) and meteorological services. The software architecture is applicable in the context of buildings. In contrast, Brundu et al. [58] presented DIMCloud, which was designed at the district level. Isikdag [5] focused on the design pattern in which GIS is applied as a base to represent and

visualize the integrated data. Peng et al. [59] proposed a risk management system based on the integration of BIM, GIS and IoT, aiming to realize real-time factor identification and effective and efficient response towards an emergency. Construction is always a complex management task that might be divided into numerous subtasks involving various actors and heterogeneous information. Huang et al. [61] proposed a BIM-GIS-IoT-based excavated soil recycling system that had an elaborated software constitution. They verified the system within a practical excavation project. To handle schedule deviation risks, Lee

Table 4

Concepts of BIM-GIS Integration Ontology and corresponding concepts in IFC and CityGML.

BIM-GIS Integration Ontology concepts	IFC concepts	CityGML concepts
Facility	IfcFacility	—
FacilityPart	IfcFacilityPart	—
Bridge	IfcBridge	Bridge
Road	IfcRoad	Road
Railway	IfcRailway	Railway
Vegetation	—	Vegetation
CityFurniture	IfcFurniture	CityFurniture
Site	IfcSite	—
Building	IfcBuilding	_AbstractBuilding
BuildingStorey	IfcBuildingStorey	—
Space	IfcSpace	Room
Element	IfcElement	—
BuildingElement	IfcBuildingElement	—
Wall	IfcWall	—
ExteriorWall	IfcWall	—
InteriorWall	IfcWall	—
Roof	IfcRoof	—
Slab	IfcSlab	—
Ground	IfcSlab	—
BuildingInstallation	—	BuildingInstallation
Beam	IfcBeam	BuildingInstallation
Column	IfcColumn	BuildingInstallation
Stair	IfcStair	BuildingInstallation
Railing	IfcRailing	BuildingInstallation
Ramp	IfcRamp	BuildingInstallation
Furniture	IfcFurniture	BuildingFurniture
Door	IfcDoor	Door
Window	IfcWindow	Window
Opening	IfcOpeningElement	_Opening
Geometry	IfcRepresentation	—
BoundarySurface	—	_BoundarySurface
ExteriorWallSurface	—	ExteriorWallSurface
InteriorWallSurface	—	InteriorWallSurface
RoofSurface	—	RoofSurface
FloorSurface	—	FloorSurface
CeilingSurface	—	CeilingSurface
GroundSurface	—	GroundSurface
boundedBy	IfcRelSpaceBoundary	boundedBy
hasGeometry	—	—
has	IfcRelContainedInSpatialStructure	—
	IfcRelAggregates	—
	IfcRelFillsElement	—
	IfcRelVoidsElement	—

et al. [62] developed a digital twin framework for real-time logistics simulation, which can predict potential logistics risks and accurate module arrival time.

In addition to the software architecture, the construction of a data lake targeting a multismart building environment is also critical in the future of urban asset management [64]. Kanak et al. [65] proposed a method to extend BIM to urban semantic context for data-driven crisis preparedness. They considered the fusion of BIM and GIS predominant data models to provide multilevel visualization, while sensor data and asset data are separately managed.

In terms of content, the studies mentioned above have a common characteristic that they all focused on presenting a design pattern of the BIM-GIS-IoT-integrated software architecture. However, in these studies, integration of the underlying data models is rarely mentioned. Traditional BIM + GIS + IoT integration methods based on software platform, most of the work is only to achieve spatial overlap and visual

integration, while the underlying data structure is still isolated, which causes obstacles to many indoor and outdoor integration applications. For example, take indoor and outdoor integrated evacuation and navigation as an example. If BIM's indoor route cannot be combined with outdoor road network, unified route planning, navigation and flow optimization cannot be realized, and sensors and devices only have spatial location information, but cannot establish topological association with spatial objects such as rooms and road networks. Therefore, they cannot automatically realize spatial positioning, status recognition and data analysis and other operations. All of these shortcomings pose an underlying, critical obstacle to the implementation of Digital Twin. Hence, it spawns a demand to present an approach to fully integrate the underlying data models of BIM, GIS and IoT.

3. Methodology

The overall framework created in this study to integrate heterogeneous BIM, GIS and IoT data is illustrated in Fig. 1. This framework has a tiered structure. The first tier is BIM-GIS integration, during which three procedures are conducted, including ontology construction, data instantiation, and ontology mapping. The geometry of the built asset is processed to meet the need of multiscale visualization, and other information from IFC, CityGML and shapefile are transformed into RDF. The first tier will form the static physical space-time container of CIM to carry all kinds of spatial entities and properties.

The second tier is BIM-GIS and IoT integration. After the construction of the BIM-GIS integrated ontology, the dynamic data will be linked to the kernel city model by binding the IoT devices and its dynamic observations. In this case, Brick and SSN are selected as the IoT source ontologies. Finally, the complete integrated domain ontology for urban management is deduced. For further applications, various application ontologies can be derived from the original integrated domain ontology to adapt different application contexts. The most essential step is to build a CIM Ontology merging heterogeneous BIM, GIS and IoT data into an urban semantic knowledge base for urban management and data sharing. From the perspective of urban data management through semantics, the proposed approach established a CIM ontology for the complete and consistent semantic expression of urban information. Both spatial and semantic information are organized by ontologies to make coherences between different levels of details.

Thus, a predefined CIM ontology framework model (named as OntoCIM) was proposed in this paper (as shown in Fig. 2). It is intended to cover and organize BIM-GIS-IoT concepts in a harmonized object-oriented semantic structure. In particular, the structure considers the multiscale resolutions of city entity objects and the coherence of both geometric and semantic dimensions. Users can be public users, researchers or government administrators, enabling them to make semantic queries based on a wide range of urban knowledge. Specifically, the following points need to be completed: 1) define coincident semantics for similar concepts; 2) change the redundant data structures into a flatter graph model and extract the “implicit” information from deeper layers to improve query efficiency; and 3) redistrict the LoD of BIM data according to the granularity of GIS information organization.

3.1. First tier: BIM-GIS integration

3.1.1. Ontology construction

The ontology modelling frameworks and principles have been

Table 5

Examples of SWRL mapping rules for ontology mapping.

No.	Content	Mapping rules
MR1	Direct mapping of concepts	<code>ifcowl:IfcBuilding(?x) -> bg:Building(?x) ifcowl:IfcWallStandardCase(?x) -> bg:Wall(?x) citygml:Room(?x) -> bg:Space(?x)</code>
MR2	Conditional mapping of concepts	<code>ifcowl:IfcSlab(?x) ifcowl:FamilyName.Pset.Ifslab(?x, "Foundation Slab"^^xsd:string) -> bg:Ground(?x)</code>
MR3	Mapping of properties	<code>ifcowl:IfcSpace(?x) ifcowl:Name.Pset.Ifspace(?x, "bedroom"^^xsd:string) -> bg:Space(?x) bg:function(?x, "bedroom"^^xsd:string)</code>

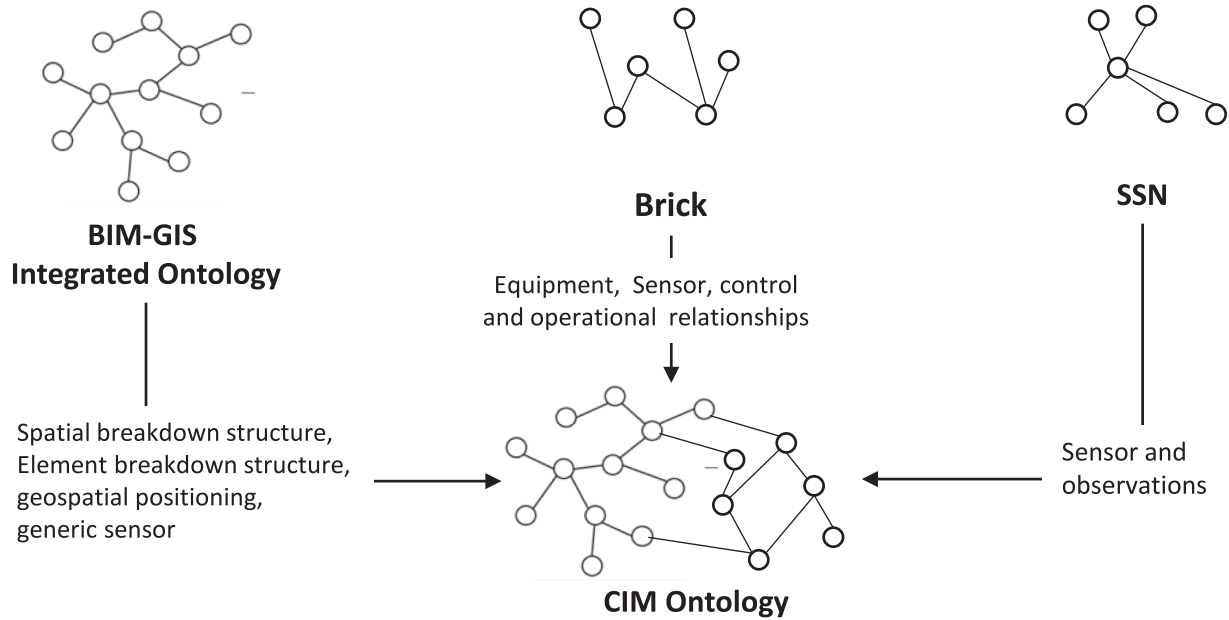


Fig. 15. The approach of the fusion of IoT and BIM-GIS integrated ontology and further applications.

Table 6

Examples of SWRL mapping rules for ontology alignment of IoT and BIM-GIS integrated ontology.

```

bg:Building(?x) -> cim:Building(?x)
bg:Wall(?x) -> cim:Wall(?x)
bg:Space(?x) -> cim:Space(?x)
bg:Sensor(?x) -> cim:Sensor(?x)
brick:Equipment(?x) -> cim:Equipment(?x)
sosa:Sensor(?x) -> cim:Sensor(?x)
sosa:Observation(?x) -> cim:Observation(?x)

```

discussed by several studies [66–68], and some domain ontology development methodologies have been proposed [67,69]. Generally, the ontology construction in this study includes the four following steps: determine the purpose and scope, define the classes and the class hierarchy, define properties and facets, and define instances.

At the highest level of abstraction, the BIM-GIS Integration Ontology contains four top main classes, namely, *Geometry*, *Element*, *Facility*, and *Feature*. Linkages between them are illustrated in Fig. 3. The “*Feature-Facility-Element*” relations correspond to low to high precisions and are linked with *Geometry* via *hasGeometry*.

The hierarchy structure of *Geometry* (see Fig. 4) is built based on the geometry module of CityGML, including all the basic geometric concepts and relations, such as *Solid*, *Surface*, *Curve*, *Point*, and *MultiGeometry*. To adapt to different spatial scales, geometric entities must have explicit expressions and LoD attributes (i.e., *lodType*). In addition, the representation literal of *Geometry* must be specified via the *asWKT* and *asGML* properties, which indicate that the geometry information is represented as Well Known Text (WKT) and Geography Markup Language (GML), respectively.

Element mainly focuses on the interior entities inside the building, which are divided into physical components and spatial components, as shown in Fig. 5. Entities of the former are mapped from *IfcElement* in

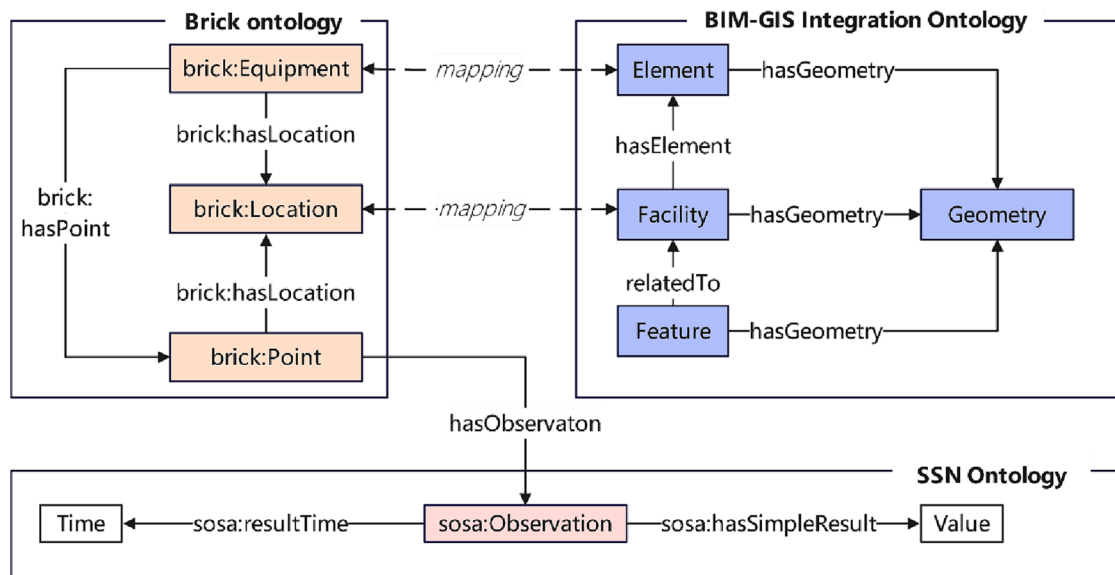


Fig. 16. The approach of the integration of BIM-GIS, BRICK, and SSN ontology.

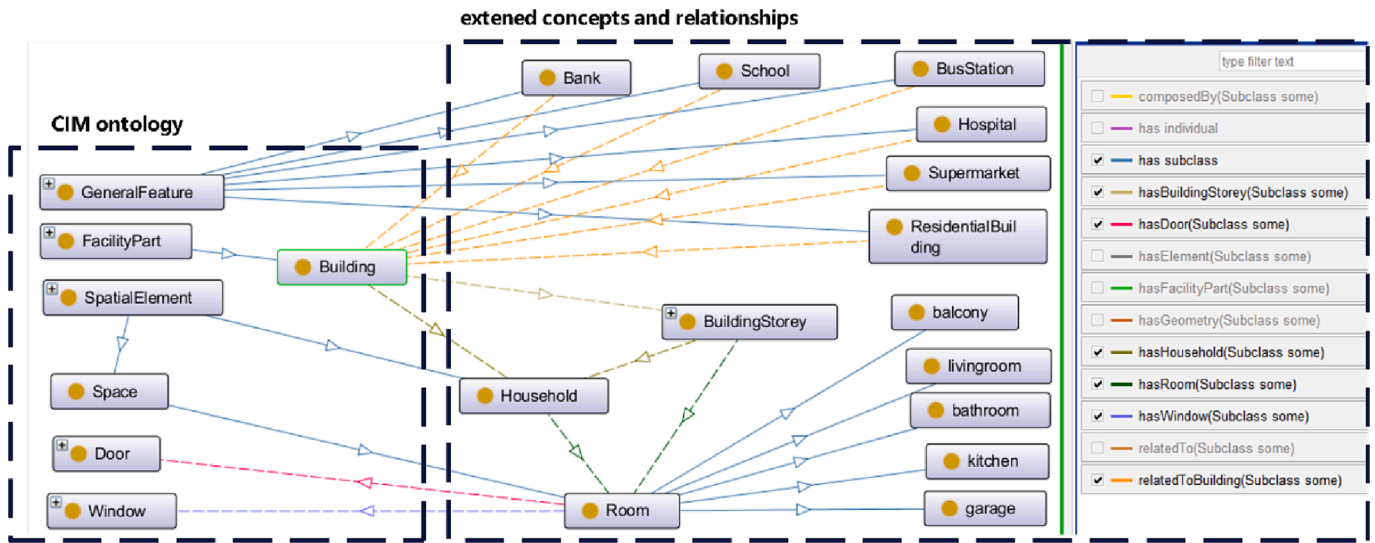


Fig. 17. Application ontology for house searching and assessment.

Table 7

Examples of mapping rules from CIM ontology and application ontology.

No.	Content	Mapping rules
MR4	Decomposition of concepts	$(?x \text{ rdf:type } \text{cim:Space})(?x \text{ cim:Name_Pset_IfcSpace } \text{"balcony"}) \rightarrow (?x \text{ rdf:type } \text{app:balcony})$
MR5	Supplementation of datatype properties	$(?x \text{ rdf:type } \text{cim:Space})(?x \text{ cim:Area_Pset_IfcSpace } ?v) \rightarrow (?x \text{ app:Area_room } ?v)$
MR6	Supplementation of object properties	$(?x \text{ rdf:type } \text{cim:Building})(?y \text{ rdf:type } \text{app:BuildingStorey})(?x \text{ cim:has } ?y) \rightarrow (?x \text{ app:hasBuildingStorey } ?y)$

IFC and LoD4 objects in CityGML (such as BuildingInstallation and BuildingFurniture), while the entities of the latter are obtained from IfcSpatialStructureElement in IFC (such as IfcBuildingStorey and IfcSpace) and spatially related objects in CityGML (such as Room and BoundarySurface). Relationships are also defined between the entities, such as *has* describing the spatial compositional relations and *boundedBy* describing the spatial boundary relations.

Facility focuses on external structures, such as a building represented as a bounding box at a lower level of granularity, such as LoD2 or LoD3, and it needs to record address and location attributes. A *Facility* is the aggregation of several *FacilityParts*, as shown in Fig. 6. For example, a *ShoppingMall* consists of two *Buildings* and a *Sites*.

Feature is mainly used to record the public use and administrative planning with management attributes and location information, including general features, land features, and shape features, as shown in Fig. 7. The geometry of the shape feature can be expressed by 2D vector data.

In addition, the datatype properties include some basic attributes, such as *id*, *name*, *description*, *function*, and *address* (see Figs. 4–7). All the other thematic properties are recorded as instances of subproperties of the “_pset” property, which are mainly extracted from IFC elements. Since IFC already has all-sided definitions according to different applications, the properties are directly mapped for subsequent use.

3.1.2. Data instantiation

Data instantiation refers to the extraction of different data sources and the transformation into a unified knowledge model. In this paper, the data from various data sources (e.g., IFC, CityGML, and shapefile) are converted into an RDF data model, which can be regarded as a

directional graph for linked data. As such, the conversion includes the node creation (i.e., the conversion of entities) and edge creation (i.e., the conversion of relationships).

3.1.2.1. IFC to RDF. The conversion from IFC to RDF is conducted based on the transformation of the IFC EXPRESS schema. An existing tool, namely, the IFC-to-RDF converter [41], has been proposed to achieve such transformation, and the output is called ifcOWL. However, ifcOWL is completely based on the IFC schema. It is redundant with multiple objects linked by complex multilevel referenced entities, so the data are at a deep level. This is contrary to the ontology’s intention to flatten a hierarchical structure. In addition, ifcOWL lacks a multiscale description of geometry. As such, converting IFC to RDF based on the proposed BIM-GIS Integration Ontology involves two aspects: the conversion of IFC objects and relationships and the geometry processing of IFC objects.

The multilevel referenced paths between IFC objects are minimized by reorganization based on ifcOWL to achieve the conversion of IFC objects and relationships. SPARQL is used to define the query path of the ifcOWL and locate the target class or property (Fig. 8). Examples of the SPARQL queries used are listed in Table 2. The query SQ1 is used to obtain the attribute of IFC objects, and the variables, *?obj*, *?n*, and *?value*, refer to an IFC object, the attribute name, and the related nominal value. These three variables consist of a triple in the RDF data model, and the attribute name is stored as a datatype property (see Fig. 8(a)). SQ2–5 are used to create spatial compositional relationships between IFC objects. Each pair of obtained IFC objects is linked according to the corresponding object properties defined in the BIM-GIS Integration Ontology (see Fig. 8(b)). For example, the entities of IfcWallStandardCase and related IfcWindow retrieved by SQ5, which are indexed as the variables? *wall* and *?dw*, modelled as the triple (?*wall*, *has*, ?*dw*) where the relationship is predefined, as shown in Fig. 5. An example of the instantiation results from IFC data is shown in Fig. 9.

Geometry definition is a fundamental part of visualization and further geometry-related analysis. IFC represents the highest precision and has a variety of geometric constructs. However, loading the most detailed geometry at one time is not conducive to practical application and efficient visualization. In comparison, GIS covers a larger spatial scale, and the multiscale geometric expression of CityGML is more suitable for various applications within the city scope. Therefore, the geometric information of IFC is first converted into B-Rep (Boundary Representation), which is consistent with CityGML. To address the

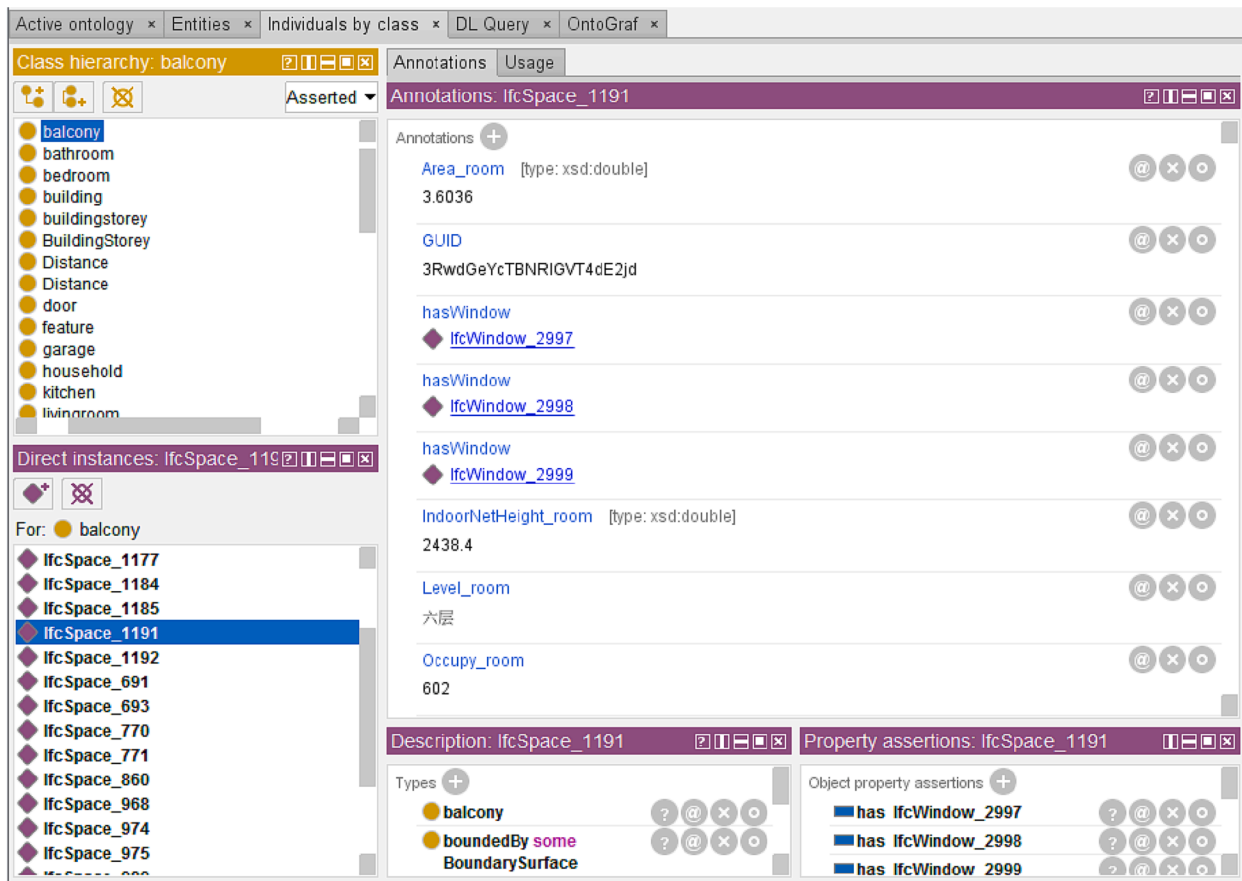


Fig. 18. Generation of new knowledge after ontology mapping.

differences in the coordination system and the geometric representation between IFC and CityGML, this study uses the FME tool to finish the transformation work.

Next, a separation of the IFC objects according to different LoDs is conducted, as listed in Table 3. LoD1 is a block-shaped model of a building with a horizontal roof. LoD2 adds a generalized roof and accessory installations such as balconies. LoD3 denotes architectural models with external walls and roofs, potentially including doors and windows. LoD4 adds interior structures for buildings such as rooms, interior doors, stairs, and furniture. During this process, the alpha-shape algorithm is applied in this study to extract geometric features and distinguish between exterior and interior walls. As shown in Fig. 10, for an IFC object, its B-Rep geometric representations of different LoDs are expressed as WKT literal or GML literal and are organized in the RDF based on the BIM-GIS integration ontology as mentioned in section 3.1.1.

3.1.2.2. CityGML to RDF. CityGML should be parsed first, and the labels and contents of all nodes in the tree-like structure should be extracted. Then, the classes or attributes to be mapped should be determined according to the meaning of nodes. Fig. 11 shows the structures of the CityGML and RDF models. It can be seen that the main content of the mapping is the transformation of elements of the tree structure and elements of attributes to the nodes and edges of the RDF model. As shown in Fig. 12, it is possible to convert elements with an id value in CityGML into class nodes in RDF and elements without values into object properties. The attributes are converted into datatype properties, and the corresponding attribute values are converted into text. An example of the conversion result is shown in Fig. 13.

3.1.2.3. Shapefile to RDF. Shapefile is a common data format developed by ESRI to describe simple nontopological vector graphics of 2D maps in a GIS system. The geometric elements are mapped to three classes, Point, Polyline or Polygon, and recorded in WKT Literal format. Shapefile dBase files record the attributes of vector objects. In a dBase table, a row represents an instance of a ShpFeature class, and the header of each column is mapped to a datatype property. The instantiation of the shapefile object is depicted in Fig. 14.

3.1.3. Ontology mapping

The above step has achieved the unification of heterogeneous data sources by representing them as an RDF data model in syntax. Nevertheless, semantic differences, such as terminology differences, remain to be solved. For example, the concept of “Building” is entitled “IfcBuilding” in IFC and “AbstractBuilding” in CityGML. In this regard, ontology mapping is conducted to address such problems. The mapping rules are created as SWRL (Semantic Web Rule Language) rules. SWRL is a rule language based on Datalog and can formulate rules by combining ontology construction (description logic) and Horn logic. SWRL allows users to write OWL-based rules to provide stronger knowledge representation and more powerful deductive reasoning capabilities. Here, the prefixes *ifcowl*, *citygml*, and *bg* represent the namespaces of IFC, CityGML, and the BIM-GIS Integration Ontology, respectively.

Basically, ontology mapping can be divided into mapping between concepts and between properties. The mapping between concepts includes direct mapping and conditional mapping. The direct mapping addresses the concepts in the two ontologies that are declared to be equivalent. For example, *IfcBuildingElement* of IFC is mapped to *BuildingElement* of the BIM-GIS Integration Ontology, and *Room* of CityGML

Query:

```
SELECT ?s ?o WHERE { ?s app:Owner "602"^^String . ?s app:Area_room ?o . }
```

Searching Results:

```
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_991> "3.2922"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_780> "12.1836"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_422> "16.5486"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_366> "8.8288"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_1357> "13.0356"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_1191> "3.6036"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#IfcSpace_1190> "5.0976"^^<http://www.w3.org/2001/XMLSchema#double>
```

a) Query of apartment 602 and areas of all rooms

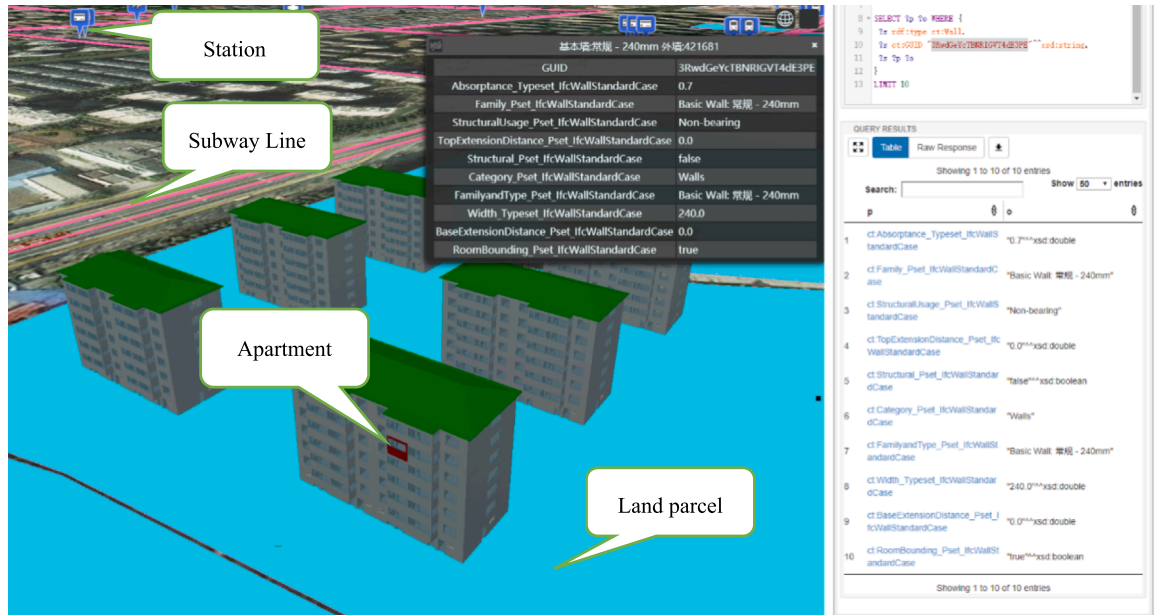
Query:

```
SELECT ?s ?o WHERE { ?s rdf:type bgip:polyfeature. ?s cim:name "Qu Yang Middle School". ?s cim:hasGeometry ?g. ?g cim:asWKT ?o. }
```

Searching Results:

```
<http://www.semanticweb.org/ontologies/cim_onto#polyfeatureB00150BCA6> "POLYGON((121.49172745168211 31.28261468788515, 121.49175814568999 31.281833887801935, 121.49176583886148 31.28142701268196, 121.4915055090189 31.28141670525074, 121.49151319146156 31.281105958223343, 121.49106144309844 31.28111414790154, 121.49106907248498 31.28143828511238, 121.49072452048726 31.281439167857172, 121.49073214471342 31.28182241022587, 121.49115326464572 31.281829380989077, 121.49116085231306 31.282611694117165, 121.49172745168211 31.28261468788515))"
```

b) Query of GIS information, i.e. searching for the school close to Room 602



c) 3D visualization of the query results: indoor and outdoor information of room 602

Fig. 19. Information of CIM instance SPARQL query results.

is mapped to *Space*, which is combined with the concept of *IfcSpace*. Some mapping concepts are listed in Table 4. The mapping rules MR1 in Table 5 are executed to perform the direct mapping. Conditional mapping requires the concepts to have indicative attributes with which the concepts can be mapped to related classes in the BIM-GIS Integration Ontology. For instance, *IfcSlab* with *FamilyName_Pset* attribute *Foundation Slab* is defined as the *Ground* in the BIM-GIS Integration Ontology, as depicted by MR2 in Table 5. These indicative attributes can also be used for the mapping of properties, such as rule MR3, which declares the *function* of a *Space*.

3.2. Second Tier: BIM-GIS-IoT integration

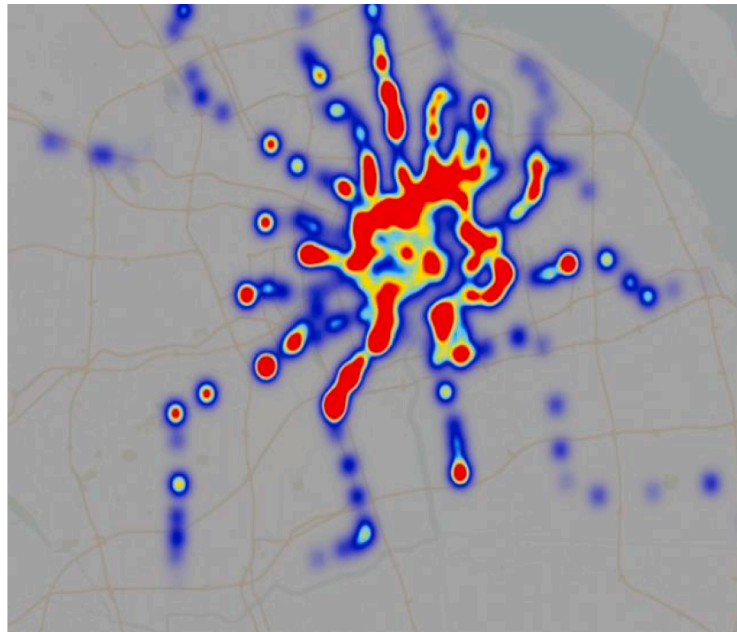
After the first tier of integration of BIM and GIS, the backbone of urban asset management has been established. More specifically, it unifies the description of the spatial breakdown structure and element breakdown structure of urban assets, which makes it possible to access the specific asset semantic data from different scales with different levels of detail.

Based on BIM-GIS integration, Brick and SSN are then introduced to gain further BIM-GIS-IoT integration. Although IFC and CityGML have

Table 8

Inference rules used to construct new facts in the application ontology.

No.	Content	Mapping rules
IR1	Define the class “large_livingroom”	(?lr rdf:type app:livingroom) (?lr app:Area_room?a) greaterThan(?a 20)-> (?lr rdf:type app:large_livingroom)
IR2	Define the class “school_nearby_building”	(?a rdf:type app:feature) (?a rdf:type app:ResidentialBuilding) (?b rdf:type app:School) (?x rdf:type app:Distance) (?x app:distRelated?a) (?x app:distRelating?b) (?x app:distValue?v) lessThan(?v 1000)-> (?a rdf:type app:school_nearby_building)
IR3	Define the datatype property “advantage” and the property value “convenient transportation”	(?a rdf:type app:feature) (?a rdf:type app:ResidentialBuilding) (?b rdf:type app:BusStation) (?x rdf:type app:Distance) (?x app:distRelated?a) (?x app:distRelating?b) (?x app:distValue?v) lessThan(?v 500)-> (?a app:advantage “convenient transportation”^^xsd:string)
IR4	Define the class “which_I_prefer” representing the target apartment	(?x rdf:type app:household) (?x app:total_area_house?a) greaterThan(?a 90) (?x app:hasRoom?r) (?r rdf:type app:large_livingroom) (?y rdf:type app:ResidentialBuilding) (?y app:relatedHouse?x) (?y rdf:type app:school_nearby_building) (?y app:advantage “convenient transportation”^^xsd:string)-> (?x rdf:type app:which_I_prefer)



a) The convenient transportation query according to the road network traffic accessibility derived from GPS data of vehicles over the last month

Query:

```
SELECT ?s ?o WHERE { ?s rdf:type app:which_I_prefer. ?s app:total_area_house ?o. }
```

Searching Results:

<http://www.semanticweb.org/ontologies/cim_onto#house201>	"95.334"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house101>	"96.4604"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house301>	"99.1644"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house204>	"95.334"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house604>	"99.1644"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house601>	"97.35000000000001"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house501>	"97.476"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house304>	"97.35000000000001"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house401>	"96.552"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house404>	"97.476"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house504>	"96.552"^^<http://www.w3.org/2001/XMLSchema#double>
<http://www.semanticweb.org/ontologies/cim_onto#house104>	"94.4604"^^<http://www.w3.org/2001/XMLSchema#double>

b) “What I prefer?” Query results by apartment -seeking rules

Fig. 20. A favorite housing query system case based on CIM ontology.

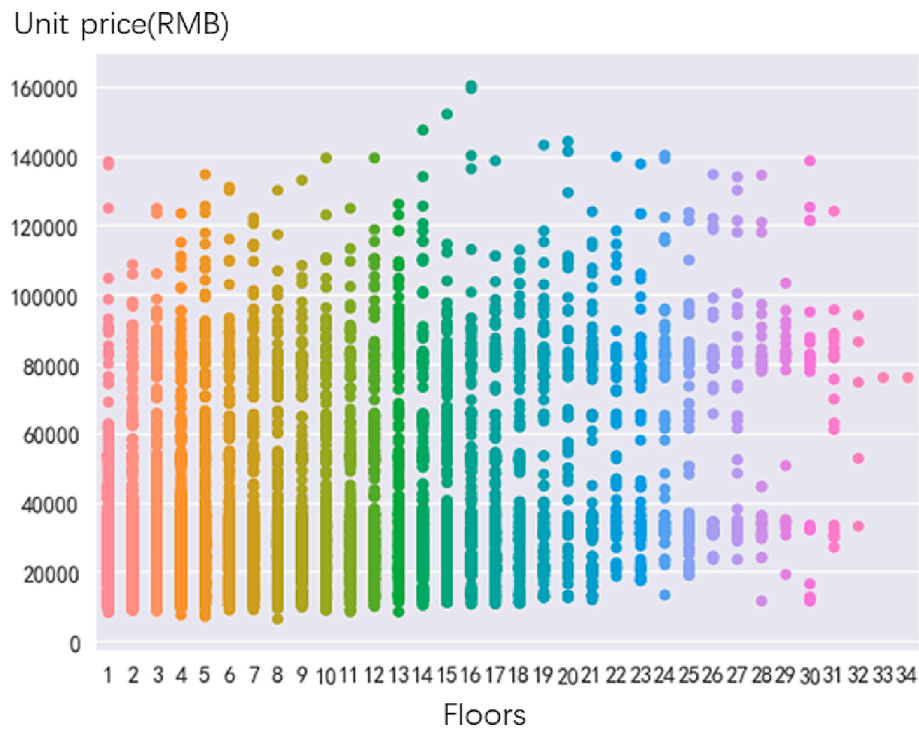


Fig. 21. The price of apartments on different floors.

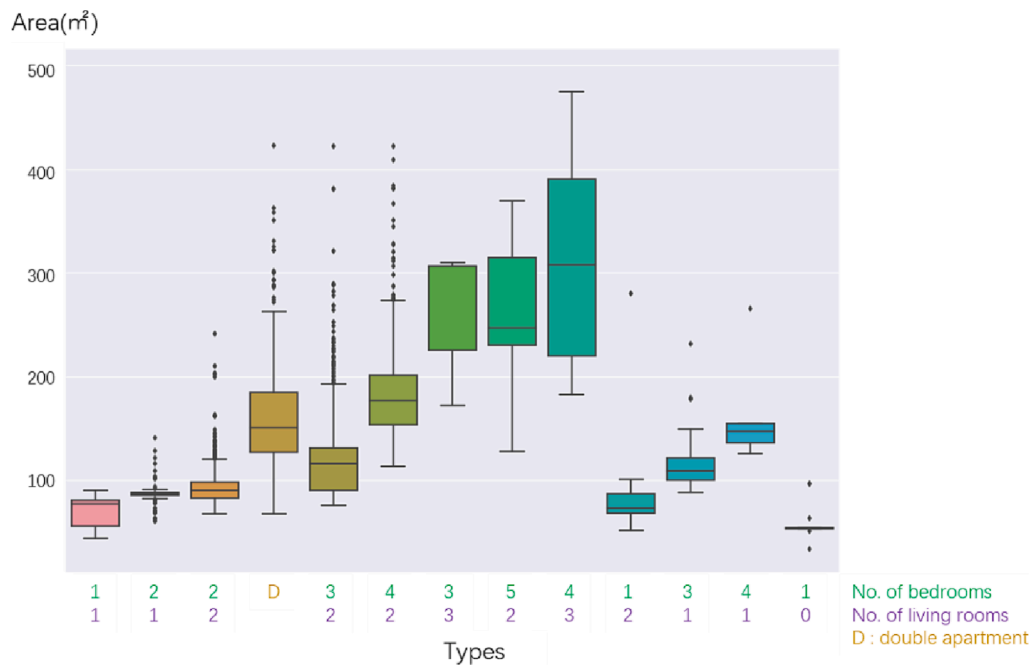


Fig. 22. The relationship between apartment types and areas.

particular data structures to capture the system breakdown of urban assets to some extent, they still have limits on modelling these concepts since they are not specialized for service systems. For the last few years, Brick has shown its power in describing the physical, logical and virtual assets in buildings and the relationships between them, while SSN has developed to be excelled in describing sensors and their observations, the involved procedures, the studied features of interest, the samples used to do so, and the observed properties, as well as actuators. It can then be generalized to the urban scale to establish a comprehensive ontology base to realize semantic integration of BIM, GIS and IoT.

Starting in 2015 [70], Brick has been one of the main open-source initiatives trying to standardize semantic descriptions of the physical, logical and virtual assets in buildings and the relationships between them [71]. By adopting semantic web technologies to construct the formal description of building automation system (BAS), it is capable to represent the building assets, including HVAC Systems, Lighting Systems, Electrical Systems, Spatial Information, Sensor Systems, and their control and operational relationships, in a flexible and linked way. As a result, it can enhance the intelligence of the BAS to support more complex applications of smart building, such as occupancy modeling

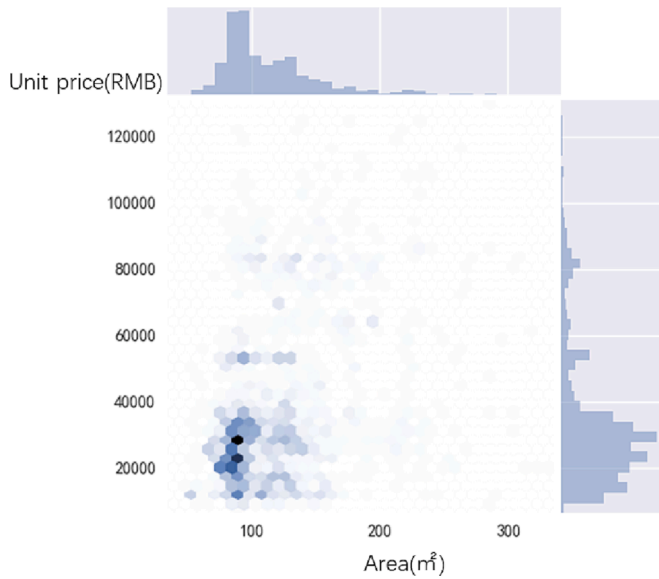


Fig. 23. Unit price of apartments - area.

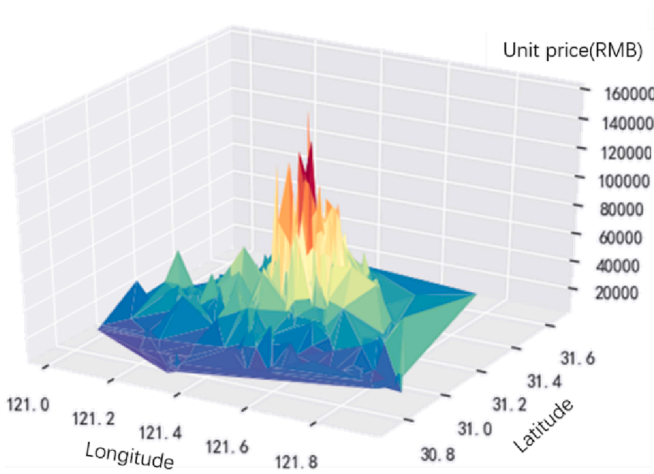


Fig. 24. Unit price distribution of apartments in case area according to CIM ontology.

[72], energy apportionment [73], Model-Predictive Control [74], Fault Detection and Diagnosis [75], Demand-Response [76] and etc.

The Semantic Sensor Network (SSN) ontology has a longer development history. It is an ontology for describing sensors and their observations, the involved procedures, the studied features of interest, the samples used to do so, and the observed properties, as well as actuators [77]. SSN was initiated by W3C semantic sensor network incubator group (SSN-XG) in 2009 [78]. The initial SSN ontology is based on the precedent work of OGC's Sensor Web, namely SensorML [79] and OMXML [80]. In addition, it is built on an ontology design pattern which is called the Stimulus Sensor Observation (SSO) pattern [81]. During the past few years, the SSN ontology has experienced a revolutionary upgrade due to the urgent demand for lightweight ontology for practical applications. Some shortcomings of the initial SSN have been addressed. The new SSN expands SSO into Sensor, Observation, Sample, and Actuator (SOSA) ontology, which serves as the new central building block for the SSN. SOSA is applicable for lightweight uses and is capable to be used standalone.

The ultimate goal of digital twin is to have the ability to simulate the behavior of the object of concern and make in-time decision based on the reasonable prediction. The foundation of this objective is to capture the characteristics and status of the target in real time and conduct the actuation orders proactively. By now, the information of building assets, sensors and their underlying relations are always encoded with different data schemas into multiple separate data silos. For the purpose of digital twin, it is critical to manage the intricate network of physical, logical, virtual assets and sensors in a more efficient way. From the perspective of completeness, ability to capture relationships and flexibility, Brick is superior to the other formal frameworks (Project Haystack, Industry Foundation Classes and Smart Appliances REference (SAREF) ontology) in describing assets in a structure and their relations [70]. On the other hand, SSN excels at describing the concept and relationships of observation and actuation. Therefore, considering the scope and capacity of description, both Brick and SSN are powerful candidates to serve as the building block of the semantic digital twin of the urban assets. The effect of integrating the ontology of IFC, Brick and SSN has preliminarily demonstrated in our previous work for fire protection [82].

From the perspective of descriptive capacity, Brick is designed to support the contents, including HVAC systems, lighting systems, electrical systems, spatial information, sensor systems, control relationships, and operational relationships, while SSN can be used to support satellite imagery, large-scale scientific monitoring, industrial and household infrastructures, social sensing, citizen science, observation-driven ontology engineering, and the Web of Things.

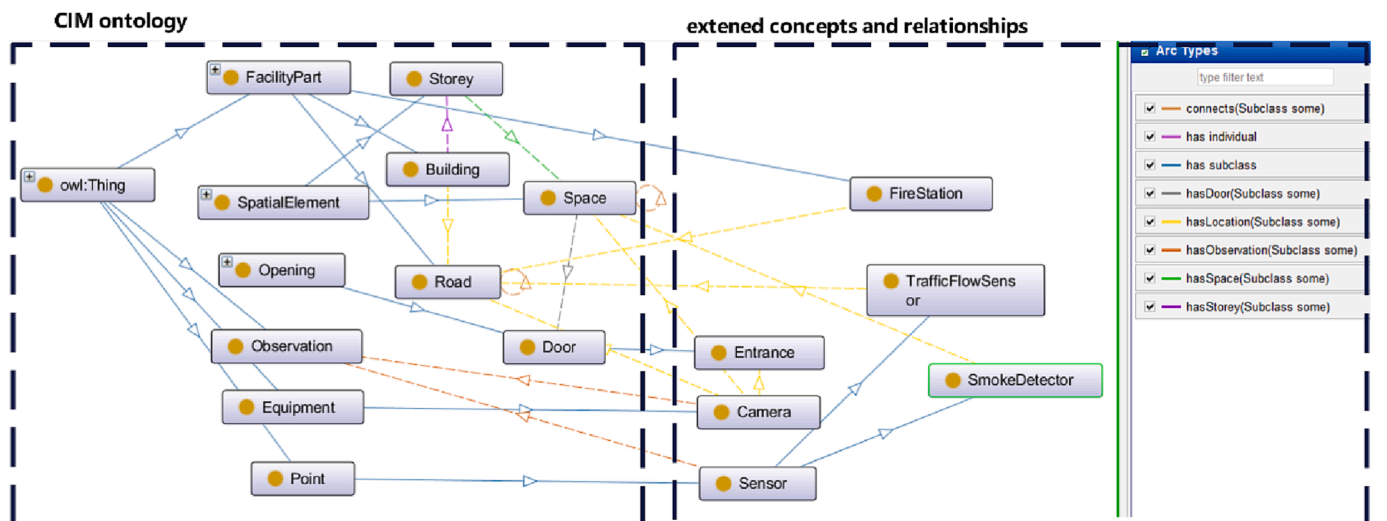


Fig. 25. Application ontology for indoor-outdoor combined evacuation strategy against fire emergency.

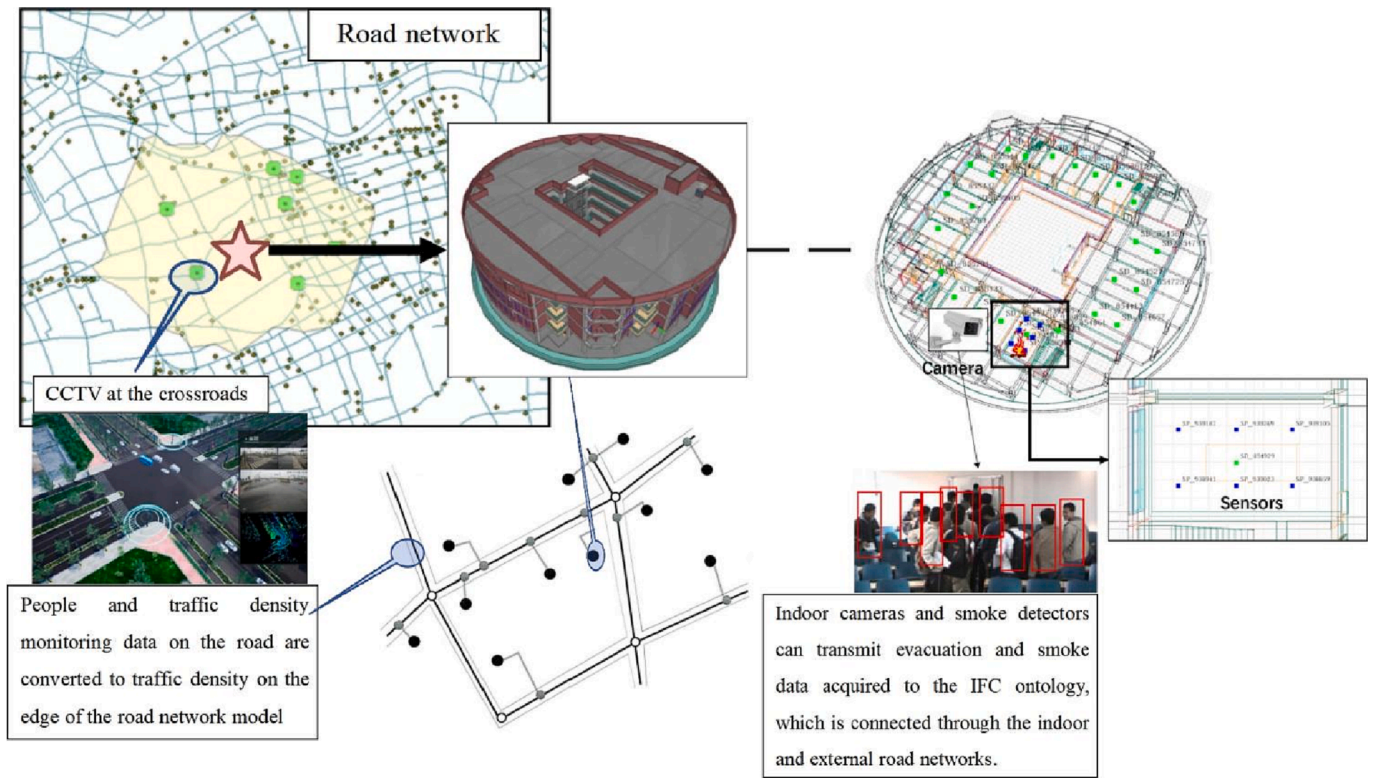


Fig. 26. A digital twin system for indoor and outdoor evacuation based on CIM ontology.

The approach of the fusion of IoT and BIM-GIS Integrated Ontology to form a CIM Ontology is shown in Fig. 15. The BIM-GIS Integration Ontology provides the geometric representation, geospatial positioning, observational relationship and additional properties of service equipment and sensors, while BRICK and SSN clarify the semantics of control relationships, operational relationships and observations. The fusion is also proceeded by mapping the concepts in BIM-GIS Integrated ontology, BRICK, and SSN ontology. The example SWRL mapping rules are shown in Table 6. Here, the prefix *cim* represents the namespaces of the proposed CIM ontology.

There are two aspect of mapping of Brick and SSN into the CIM ontology, as shown in Fig. 16. The first is mapping the spatial element of Brick (*brick:Location*) and BIM-GIS integrated ontology. The second is making the links between the sensor point, *brick:Point* from Brick, and the observation, *sosa:Observation* from SOSA.

4. Case study: Extend the CIM ontology to application ontology

The CIM ontology mainly fuses the upper concept, and a large number of datatype properties have not been well identified and utilized, so it lacks pertinence for practical application. Therefore, on the basis of CIM ontology, application ontology can be built for specific application domains. Two case studies are investigated in this section to validate the feasibility and the extension of the proposed CIM ontology, and the implementation system is developed based on Jena.

4.1. Urban assets management

This subsection takes the search and evaluation of urban assets management, such as apartment information and price, as a simple application case. First, an application ontology for housing information is established by extending necessary concepts and relationships based on CIM ontology, as shown in Fig. 17. Users may need to pay attention to both the interior of the house and the information of the surrounding environment when conducting personalized analysis and price

evaluation, so it is necessary to combine the data of BIM and GIS. The application ontology pays more attention to details and concepts in specific scenes. For example, the class “space” in the CIM ontology represents all enclosure space individuals, but it should be subdivided into bedrooms, kitchens and other subclasses according to their specific functional types in the application ontology.

The extension from CIM ontology to application ontology emphasizes on the decomposed mapping of concepts and the supplementation of properties, which could be realized by mapping rules lining with Jena rule syntax. Examples of the mapping rules are listed in Table 7. In order to achieve decomposition of concepts in the CIM ontology, related attributes are required. For example, the Jena rule MR4 is utilized to map the concept *Space* in CIM ontology with the concept *balcony* in the application ontology. Here, the prefixes *cim* and *app* represent the concepts in the CIM ontology and the application ontology, respectively. Besides, the name attribute of the *Space* is necessary for completing the mapping process. The supplementation of properties involves the data-type properties and the object properties. For instance. The mapping rule MR5 is used to supplement the area of a room in the application ontology, and the mapping rule MR6 is conducted to create relationship between *Building* and *BuildingStorey*. The mapping results are shown in Fig. 18, where new classes (e.g., *balcony*) and new properties (e.g., *Area_room*) are generated for the room in the application ontology.

On this basis, the semantic advantage of Ontology can be used to establish new semantic labels based on user requirements and filtering conditions. These requirements are converted into SPARQL queries to retrieve target information. As introduced above, SPARQL is an RDF-specific query language that queries information by graph patterns and rule-like conditions. Its language structure is divided into three main parts: the SELECT clause for defining the target variable, the WHERE clause for defining the search conditions, and the FROM clause for specifying the data source. Jena first loads the ontology knowledge model as an OntModel and then executes the input SPARQL query clauses by invoking QueryExecutionFactory. Some query results are illustrated in Fig. 19.

Complex requirements can also be conducted by using a set of inference rules before searching. It is assumed that the screening criteria of users are as follows: 1) the total house area is no less than 90 m²; 2) a large living room is needed; 3) the house is a school district; and 4) convenient transportation. Accordingly, the rules in Table 8 can establish these requirements to construct new facts in the application ontology. These facts can be obtained by a simple SPARQL rule, as shown in Fig. 20, and the inference processes occur once for all. When other users have the same requirements, only the query procedure is executed.

The advantage of ontology-based query is that users do not need to know the structures of BIM and GIS data sources and can access them semantically. Moreover, the data are mutually linked, different from the key-value structure of a relational database, making it possible to explore new relations. The semantics and logics also allow many pieces of information to be inferred by rules, even if they do not exist in the original data source. Currently, the experimental data are relatively limited. If the data scope can be expanded to the city level, a flexible semantically linked integrated data network will be formed. At the same time, rule-based reasoning can be used to continuously expand the urban knowledge model to continuously expand new information from limited original data.

When it comes to evaluating development value of some apartment, to query and analyze some critical influence factors could be a good choice in terms of query result above mentioned. Now that factors influencing apartment price are various and complicated, we can hardly describe it without reasonable models and fine and sufficient information. Fortunately, historical data of the real estate prices can be found easily from the Internet.

There are various types of factors which impact new apartments' prices, especially the characteristics of the apartment itself (such as floors, area, and apartment types) as well as surrounding supporting facilities (e.g. subway stations, schools, hospitals and commercial centers).

The characteristics of apartments is mainly derived from BIM data. Taking floors of apartments as an instance, Fig. 21 shows the price of apartments located on different floors. In addition, the relationship between apartment types and areas were counted in this district according to the BIM data and GIS buffer analysis, such as the area of apartments with one bedroom and no living room are usually less than 50 m² (as shown in Fig. 22). Fig. 23 gives relationship between price of apartments and apartments' area. The result shows the most popular area of apartment is about 90 m² to 100 m². These two legends explain the gradation of color in Fig. 23 together, from which we can infer that developers design area, types and the number of new apartments in order to gain more profits.

If we prefer to know the original value of apartment before developers' intervention, it is vital necessary to find the connection between apartments' unite price and geographic coordinate. Fig. 24 presents the statistic unit price of apartments at different location. Due to the fact that adjoining spaces share the same surrounding supporting facilities and similar policy, the value of apartment shows obvious geographical correlation.

4.2. Indoor-outdoor combined evacuation strategy against fire emergency

Fire protection is an essential topic in urban public safety management. Rapid response and efficient evacuation strategy against fire emergency is of significance to reduce loss of life and property. The integration of BIM, GIS, and IoT can facilitate an indoor-outdoor combined evacuation strategy and thus results in more precise rescue dispatch. On the one hand, the integration of BIM and IoT can provide fire scene information and spatial topology of a building for indoor evacuation. On the other hand, the fusion of BIM model, road network and real-time crowd detection and traffic condition distinguishing is considered as the basis of the decision making of outdoor evacuation.

A prototype of the application ontology for indoor-outdoor combination against fire emergency is designed to describe the integration of the above information, as shown in Fig. 25. Two basic facilities, i.e., *Building* and *FireStation*, are considered in the application ontology. The spatial topologies of a *Building* are specified by *Storey* and *Space* via relationships *hasStorey* and *hasSpace* respectively. The locations of these two facilities are defined by the relationship *hasLocation* and is specified as *Road*. A relationship, *connects*, is defined between two *Roads* to establish a outside road network and between two *Spaces* to establish an inside room network. Besides, two types of sensors are involved: *SmokeDetector* is used to monitor the smoke distribution within a space and sound the alarm when the smoke density exceeds the threshold and thus is linked to *Space* via the relationship *hasLocation*; and *Traffic-FlowSensor* is used to monitor the real-time traffic flow and thus is linked to *Road* via the relationship *hasLocation*. The monitoring results are stored as *Observation* via the relationship *hasObservation*. An equipment *Camera* is also defined as an IoT terminal to reflect the real-time scene information.

In order to validate the implementation of the proposed application ontology combined with the CIM ontology, this paper conducted a case study for a library using simulation techniques. The whole scene is shown in Fig. 26.

5. Conclusion

Focusing on the heterogeneity of BIM, GIS and IoT data in the process to form the CIM model of Digital Twin City, this paper uses semantic web and ontology technology to integrate the spatial and semantic information of CIM and mainly completes the following research work: 1) The methodology of data integration in different fields by using ontology is explored and realized; 2) The geometric transformation from IFC to CityGML is studied, and the unified geometry ontology of multiple LoDs is established; 3) The construction and knowledge extraction methods of BIM and GIS data source ontology are studied; 4) Constructing the multiple-scale CIM ontology for merging BIM, GIS and IoT data is realized.

The ontology-based methodology has built a general pattern for CIM data integration by mapping and linking concepts from the semantic level. It avoids changes in the original data sources and the missing data problem.

Most BIM-GIS tasks are performed in a geospatial context, but there are data requirements in different domains and at different levels of detail. Considering this problem, the CIM ontology is constructed in this paper to make a high coherence of both semantic and geometric concepts, making the structure more reasonable and interoperable. In regard to the application level, the application ontology is built, and all the application data come from the shared knowledge model of CIM ontology but are combined by a more practical structure for specific uses. Rule-based mapping is implemented between data sources, the CIM ontology, and the application ontology for data sharing within different semantic models.

There are still many deficiencies in this research. First, this established ontology is mainly focused on building-related concepts, but for urban infrastructures such as roads, bridges, pipelines, etc., the CIM ontology is not comprehensive enough and still needs to be continuously improved in future research. Second, the difficulty of using ontology technology for data integration lies in the mapping and linking process. Although automatic data instantiation is realized in this paper, it still has many inevitable processes of manual modelling and mapping. To date, there have been some studies based on artificial intelligence and machine learning methods for the automatic alignment of ontologies, knowledge extraction and relationship mining, which are expected to improve the efficiency of ontology modelling. Finally, due to the limitation of data sources and application scenarios, the further in-depth use of data is limited, and it is necessary to combine specific applications and technology to carry out deeper research and exploration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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